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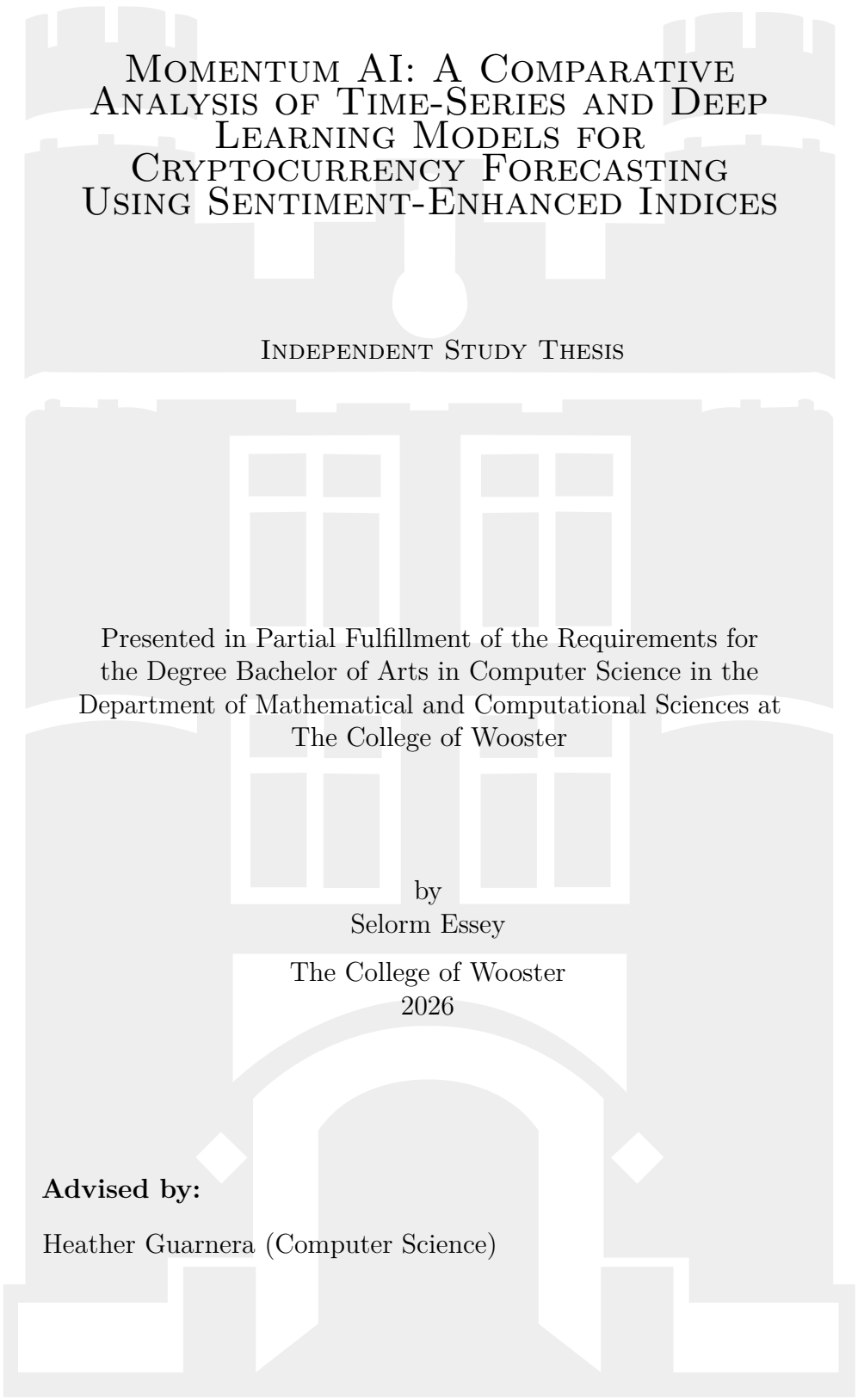
Momentum AI: A Comparative Analysis of Time-Series and Deep Learning Models for Cryptocurrency Forecasting Using Sentiment-Enhanced Indices

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MOMENTUM AI: A COMPARATIVE ANALYSIS OF TIME-SERIES AND DEEP LEARNING MODELS FOR CRYPTOCURRENCY FORECASTING USING SENTIMENT-ENHANCED INDICES

INDEPENDENT STUDY THESIS

Presented in Partial Fulfillment of the Requirements for
the Degree Bachelor of Arts in Computer Science in the
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by
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ABSTRACT

The proposed framework, Momentum AI, evaluates Naïve persistence baselines, ARIMA models, and Long Short-Term Memory (LSTM) networks under strict chronological validation and preprocessing steps designed to prevent look-ahead bias. In parallel, a transparent Momentum Fear and Greed Index (MFGI) is constructed using three explicitly defined components: price momentum, realized volatility, and aggregated daily news sentiment derived from publicly available cryptocurrency headline data. All normalization procedures rely on expanding windows to prevent look-ahead bias, and sentiment features are temporally lagged to preserve causal ordering.

Empirical results demonstrate that increasing model complexity alone does not overcome structural forecasting limitations. While nonlinear models capture short-run temporal structure, all approaches experience substantial degradation during shifts between market conditions. The inclusion of sentiment-derived features helps the framework better reflect broader market conditions and makes market behavior easier to interpret, though improvements in predictive accuracy remain conditional rather than universal.

By emphasizing transparency, careful chronological evaluation, and contextual modeling that accounts for changing market conditions, this thesis clarifies the structural limits of short-horizon prediction in digital asset markets while offering a structured framework for sentiment-aware analysis.

This work is dedicated to God.

CONTENTS

Abstract	iii
Dedication	iv
Contents	v
List of Figures	vii
List of Tables	ix
CHAPTER	PAGE
1 Introduction	1
1.1 Motivation and Problem Statement	3
1.2 Momentum AI Overview	4
1.3 Contributions	5
2 Cryptocurrency Forecasting	8
2.1 Evolution of Cryptocurrency Forecasting	8
2.2 Price-Only Forecasting	12
2.2.1 Naïve Persistence Baseline	14
2.2.2 ARIMA Forecasting	16
2.2.3 Price-Only LSTM Forecasting	19
2.3 Comparative Results Across Models (RQ1)	23
3 Sentiment in Cryptocurrency Markets	26
3.1 Role of Sentiment	27
3.2 The Alternative.me Fear and Greed Index (AFGI)	31
3.3 The Momentum Fear and Greed Index (MFGI)	33
3.3.1 Sentiment Preprocessing and Discretization	34
3.3.2 Momentum Component	40
3.3.3 Volatility Component	41
3.3.4 Sentiment Component	42
3.3.5 Composite Aggregation	43
3.3.6 Comparison with Alternative.me	46
4 Implementation and Analysis of Models	51
4.1 Price-Only LSTM	52
4.2 Hybrid LSTM with Lagged MFGI Features	56
4.3 Comparative Analysis (RQ2)	59

5	Conclusion	63
5.1	Overall Synthesis of Findings	64
5.2	Answer to RQ3	65
5.3	Limitations	67
5.4	Future Work	69
APPENDIX		PAGE
A	Use of Artificial Intelligence	71
	References	73

LIST OF FIGURES

Figure	Page
2.1 Comparison of major cryptocurrency forecasting paradigms.	10
2.2 Daily Bitcoin closing price over the past five years.	13
2.3 Naïve persistence forecast over the shared Bitcoin test window.	15
2.4 ARIMA(3, 1, 2) forecast over the shared Bitcoin test window. The dashed gray vertical line marks the chronological boundary between the training data and the held-out test window.	19
2.5 Price-only LSTM predictions versus true Bitcoin prices on the shared test window.	22
3.1 Google Trends search interest for the query “Bitcoin crash” from 2021 to 2022. Peaks in search activity reflect heightened public attention and fear during periods of market stress, illustrating the role of behavioral signals in cryptocurrency markets.	30
3.2 Example of a widely referenced Fear and Greed Index interface. While such visualizations provide intuitive summaries of market emotion, their underlying data sources, feature definitions, and weighting schemes are often undisclosed, limiting their scientific interpretability and transparency.	32
3.3 Daily mean Bitcoin news sentiment after aggregation from headline-level observations. The series remains noisy, but aggregation provides a temporally aligned daily signal suitable for downstream modeling.	35
3.4 Distribution of daily mean sentiment values after daily aggregation. Most observations lie near the center of the distribution, indicating that strongly polarized sentiment days are comparatively less common.	36
3.5 Class balance after quantile-based discretization of daily sentiment into fear (−1), neutral (0), and greed (+1) states. The classes are nearly evenly represented, supporting balanced regime-based modeling.	38
3.6 Bitcoin closing price and daily mean news sentiment aligned to the same daily calendar. The two series are related, but they do not move in lockstep, which motivates the use of sentiment as contextual information rather than as a direct proxy for price.	39
3.7 Bitcoin closing price and Momentum Fear and Greed Index (MFGI). The index summarizes broader market conditions rather than directly mirroring the raw price series.	46

3.8	Overlay of the transparent Momentum Fear and Greed Index (MFGI) and the proprietary <i>alternative.me</i> Fear and Greed Index over their overlapping historical window.	47
4.1	Price-only LSTM predictions versus actual Bitcoin closing prices over the held-out test period.	55
4.2	Hybrid LSTM predictions versus actual Bitcoin closing prices using price and lagged MFGI inputs.	58
4.3	Comparison of price-only and hybrid LSTM forecasts against actual Bitcoin closing prices. The hybrid model remains smoother than the observed series, but generally tracks the target slightly more closely than the price-only baseline.	61

LIST OF TABLES

Table	Page
2.1 Comparison of price-only forecasting performance on the shared Bitcoin evaluation window. Lower values indicate better predictive performance.	23
3.1 Example rows of lagged sentiment features. Both discretized and raw lagged sentiment values are shown, ensuring that only past information is available to later forecasting models.	40
3.2 Example rows showing the normalized momentum, volatility, and sentiment scores used to construct the MFGI, together with lagged MFGI features for later hybrid forecasting experiments.	45
4.1 Comparison of price-only and hybrid LSTM forecasting performance. Lower values indicate better predictive performance.	60

CHAPTER 1

INTRODUCTION

Cryptocurrency has become one of the most popular yet unpredictable investment opportunities of the modern financial era. It offers the possibility of enormous gains while also presenting the risk of significant losses, often within very short timeframes. For many new participants, entering the crypto market is remarkably simple. A smartphone, an internet connection, and a modest amount of capital are usually enough to begin trading on a digital exchange. This accessibility has attracted millions of investors across the world and has created a marketplace that is vibrant, experimental, and continuously evolving. At the same time, the low barrier to entry exposes inexperienced traders to a range of risks that are easy to underestimate. The cryptocurrency market operates without the structural pauses, oversight, and stabilizing mechanisms that characterize traditional financial systems. Prices can shift within seconds, and these movements are shaped not only by technological updates or regulatory announcements, but also by the collective psychology and reactions of the global trading community.

As the number of cryptocurrencies has grown, many investors have attempted to manage uncertainty by diversifying across multiple coins. The principle of not placing all one's resources into a single asset is widely embraced within the community. However, diversification in the crypto ecosystem often functions differently than it does in traditional finance. Many tokens lack stable fundamentals, clear real-world applications, or long-term viability. As a result, spreading funds across several unstable assets can increase overall risk rather than

reduce it. A portfolio containing numerous coins may still experience widespread losses if broader market sentiment turns negative. Since many assets respond to shared emotional and social influences, downturns often affect the entire market at once. Minimal regulatory oversight within the space further intensifies this unpredictability, allowing manipulation, misinformation, and rumor-driven trading to spread more easily.

The cryptocurrency market is distinct from conventional financial systems in both structure and behavior. It blends speculative trading dynamics with the influence of online communities, rapid technological innovation, and narrative-driven attention cycles. Unlike stock markets, where prices often reflect company performance, macroeconomic indicators, or long-term economic trends, cryptocurrencies are far more sensitive to shifts in public sentiment, internet culture, and emerging narratives. A trending social media post, a widely shared rumor, or a short-lived viral movement can alter billions of dollars in market capitalization. These rapid changes generate an environment that is exciting for traders who thrive on volatility. For researchers and analysts who aim to forecast price movements, however, this creates a serious challenge. Predictive models must attempt to capture not only numerical patterns in price data, but also the emotional and social forces that influence decision-making in real time.

Another defining characteristic of the crypto environment is the persistent threat of manipulation, fraud, and trust shocks. While traditional markets benefit from layers of regulation and institutional oversight, digital asset markets often rely on informal community governance and self-regulation. This freedom supports rapid innovation and experimentation, yet it also enables deceptive practices and fragile market confidence. High-profile events such as the TerraUSD collapse [13], the FTX bankruptcy [14], and major regulatory actions against exchanges [17] illustrate how quickly sentiment can reverse and how strongly prices respond to credibility-related disruptions. These events cause direct financial harm and also weaken trust within the community. Each instance reinforces the perception that the market is emotionally reactive, information-sensitive, and capable of shifting regimes abruptly.

1.1 MOTIVATION AND PROBLEM STATEMENT

Given the combination of technical complexity and emotional volatility in cryptocurrency markets, traditional forecasting tools are often not sufficient. Approaches that rely solely on historical prices overlook behavioral and psychological factors that influence how traders act collectively. In practice, many sudden price swings are not explained by gradual statistical drift; they are driven by attention, fear, hype, and shifts in perceived legitimacy. Forecasting models that fail to incorporate market mood may therefore struggle to anticipate rapid crashes, sharp rebounds, or irregular price patterns.

At the same time, many modern forecasting systems that do incorporate richer signals are difficult to evaluate transparently. Complex machine learning pipelines may improve predictive performance, but they can also become opaque: it becomes unclear *why* a model produced a specific forecast, which inputs mattered most, or whether the pipeline can be clearly examined by other researchers. In financial contexts, this opacity is not just an academic issue. When people rely on predictions to make decisions with real stakes, interpretability and transparency become part of responsible research practice.

Prior work has shown that integrating economic, technological, and machine learning features can improve predictive performance in cryptocurrency markets [6, 4]. Comparative studies also highlight important trade-offs between classical statistical models such as ARIMA and deep learning architectures such as LSTM networks [9, 10]. More recent work suggests that incorporating sentiment signals into predictive architectures can enhance forecasting accuracy, particularly in volatile financial environments [7]. Collectively, this literature supports a clear conclusion: an effective forecasting framework for cryptocurrency should not be price-only. It should integrate financial indicators with signals that reflect the emotional state of market participants, while remaining methodologically clear and transparent.

1.2 MOMENTUM AI OVERVIEW

This realization forms the foundation of *Momentum AI*. I designed *Momentum AI* as a transparent forecasting system that integrates quantitative modeling with measures of public sentiment. A central component of the project is a custom Momentum Fear and Greed Index (MFGI) that quantifies collective market emotion using openly accessible data sources and a transparent formula for computation. In this thesis, sentiment signals are derived from cryptocurrency news headlines contained in a publicly available dataset [1]. This dataset aggregates coverage from reputable media outlets (e.g., CoinDesk) and provides sentiment scores produced through natural language processing methods. Rather than treating sentiment as an abstract concept, the MFGI operationalizes it as a measurable signal that can be tracked over time and integrated into a forecasting pipeline.

The forecasting framework also incorporates a range of predictive models, allowing performance to be evaluated under consistent experimental conditions. These models include simple baselines such as the Naïve model, statistical models such as ARIMA, and deep learning models such as Long Short-Term Memory networks. The project is organized so that forecasting first begins in a price-only setting and then expands to a sentiment-aware setting through the addition of the MFGI. This makes it possible to evaluate what historical prices alone can explain before testing whether behavioral context provides meaningful added value.

The project is structured around three guiding research questions:

- **RQ1:** Which forecasting models perform best for short-term cryptocurrency price prediction when evaluated under identical experimental conditions?
- **RQ2:** Does incorporating external sentiment signals, as represented by the Momentum Fear and Greed Index, improve predictive accuracy beyond what can be achieved using historical price data alone?
- **RQ3:** How can transparency and interpretability be preserved within a forecasting pipeline that relies on complex machine learning architectures?

These research questions emphasize the project’s focus on both predictive performance and methodological clarity. In particular, the thesis treats interpretability as a design goal rather than an afterthought: the objective is not only to generate forecasts, but to provide context for understanding *when* the market is in a fear-driven regime versus a greed-driven regime, and how that regime awareness relates to model behavior. Together, the three questions also define the structure of the thesis itself: the study first examines the limits of price-only forecasting, then evaluates the contribution of a transparent behavioral signal, and finally reflects on what transparency makes possible in sentiment-aware financial modeling.

1.3 CONTRIBUTIONS

The broader aim of *Momentum AI* is to encourage a more transparent and interpretable approach to financial prediction. The cryptocurrency ecosystem is one in which misinformation spreads quickly and technical opacity can contribute to misunderstanding, overclaiming, or exploitation. A forecasting system built on open, well-documented methods enables researchers and investors to examine how predictions are generated, which features matter most, and how behavioral signals interact with market outcomes. In markets heavily influenced by narrative, speculation, and collective attention, transparent behavioral signals can serve as useful complements to conventional price-based analysis. They may not eliminate uncertainty, but they can improve regime awareness and clarify the broader environment in which price movements occur. In that sense, *Momentum AI* positions itself as both a forecasting tool and an ethical framework for studying cryptocurrency markets.

The contributions of this thesis are as follows. First, it presents a transparent framework for constructing a sentiment-based Momentum Fear and Greed Index (MFGI) using publicly available cryptocurrency news data. In doing so, it offers a more interpretable alternative to the widely used but methodologically opaque Alternative.me Fear and Greed Index (AFGI). Second, it provides a controlled empirical comparison of Naïve persistence, ARIMA, and

LSTM models for short-term cryptocurrency price forecasting under identical experimental conditions. Third, it evaluates the impact of incorporating sentiment-derived features on predictive performance relative to price-only models. Fourth, it shows that a transparent sentiment-aware index can function both as a forecasting feature and as an interpretable analytical tool for identifying broader market regimes. Finally, it emphasizes interpretability and methodological clarity, demonstrating that sentiment-aware forecasting pipelines can remain analytically useful without relying on black-box design.

This thesis also contributes to a broader methodological argument. In fields where opaque systems are often treated as unavoidable, *Momentum AI* suggests that transparency can remain central without making a project analytically weak. A transparent system may not always be the most aggressive or highly optimized option, but it can still be useful, interpretable, and methodologically credible. This argument is especially relevant in cryptocurrency settings, where opacity can easily be mistaken for sophistication. Because digital asset markets are already vulnerable to hype, manipulation, and overclaiming, transparent analytical tools may be especially valuable. They do not remove uncertainty, but they make uncertainty easier to understand, question, and evaluate.

This thesis proceeds as follows. Chapter 2 examines the evolution of cryptocurrency forecasting research and evaluates price-only forecasting through Naïve persistence, ARIMA, and LSTM models, thereby addressing the first research question. Chapter 3 introduces the behavioral finance and sentiment foundation of the project, examines the Alternative.me Fear and Greed Index (AFGI) and its limitations, and develops the framework for the transparent Momentum Fear and Greed Index (MFGI). Chapter 4 compares a price-only LSTM with a sentiment-aware LSTM that incorporates lagged MFGI features, thereby addressing whether transparent behavioral context improves forecasting beyond price history alone. Finally, Chapter 5 synthesizes the broader implications of the thesis by reflecting on transparency, interpretability, limitations, and future directions for sentiment-aware cryptocurrency forecasting.

Ultimately, this project contributes to ongoing discussions about transparency and the responsible use of artificial intelligence in financial contexts. By integrating behavioral data with algorithmic modeling in a transparent framework, *Momentum AI* demonstrates how explainable sentiment measures can complement quantitative forecasting techniques. In markets characterized by uncertainty and rapid emotional shifts, transparency strengthens interpretability, supports accountability, and promotes more informed decision-making.

CHAPTER 2

CRYPTOCURRENCY FORECASTING

This chapter examines the evolution of cryptocurrency forecasting research and then narrows to the benchmark models selected for this study: Naïve persistence, ARIMA, and LSTM. The goal is to evaluate how these price-only approaches perform on Bitcoin data before introducing sentiment-aware extensions in later chapters.

2.1 EVOLUTION OF CRYPTOCURRENCY FORECASTING

Cryptocurrency forecasting has progressed through several distinct eras, each shaped by the availability of data, developments in computational methods, and growing recognition that digital assets behave differently from traditional financial instruments [10]. Early research treated Bitcoin and similar cryptocurrencies as conventional time series, relying on linear models that assume stable and interpretable relationships across time. This approach naturally led to the use of well-established econometric tools such as Autoregressive Integrated Moving Average (ARIMA) and volatility models such as GARCH [10]. These models became common starting points because they offer mathematical transparency, clear assumptions, and explicit formulas for describing how shocks propagate, how trends evolve, and how volatility behaves under relatively stable conditions.

As more empirical data accumulated, researchers discovered that cryptocurrency markets do not conform to the structural stability these models require. Bitcoin’s price history

displays sudden regime shifts, rapid boom-and-bust cycles, extended periods of volatility clustering, and return distributions with heavy tails. These features violate the assumptions underlying traditional linear time-series models [10]. In practice, ARIMA may track short-term patterns during quiet periods, but its performance deteriorates during extreme rallies, abrupt corrections, or broader structural breaks. Even volatility-oriented models such as GARCH, although helpful for modeling heteroskedasticity, often fail to adjust quickly when markets move through unusually turbulent or speculative phases [10, 9]. These limitations encouraged researchers to search for methods that are more flexible and capable of adapting to structural changes.

As a result, forecasting research turned toward machine learning approaches. Unlike classical econometric models, machine learning techniques do not require strict assumptions about stationarity or linearity. Methods such as Support Vector Machines, Random Forests, and Gradient Boosted Trees can learn complex nonlinear interactions between variables such as price, volume, technical indicators, and sentiment-based features [6]. Their flexibility is particularly valuable in cryptocurrency markets, where the relationships between inputs and outputs evolve quickly and may not correspond to stable economic mechanisms. These models introduced the possibility of learning patterns that are not explicitly specified by theory, opening new directions for forecasting research [10].

Hybrid modeling frameworks emerged as researchers recognized that no single technique could fully capture the multifaceted behavior of cryptocurrency markets. These hybrid approaches often combine a linear model that captures broad temporal structure with a nonlinear component that models short-term deviations. For example, a model may use ARIMA to represent low-frequency dynamics while relying on a neural network to capture nonlinear patterns that remain in the residuals [4]. Ensemble strategies that average or weight predictions from multiple models gained popularity because they reduce sensitivity to noise and improve robustness across different market conditions [4].

Figure 2.1 summarizes the major forecasting paradigms used in cryptocurrency markets

Model Category	Representative Methods	Key Assumptions	Strengths	Limitations in Crypto Markets	Motivation for Next Paradigm
Linear Time Series	ARIMA	Stationarity, linear relationships	Transparent, interpretable	Breaks under regime shifts and volatility	Need for flexible nonlinear models
Volatility Models	GARCH	Stable volatility dynamics	Models heteroskedasticity	Slow adaptation in extreme markets	Shift toward ML approaches
Machine Learning	Random Forest, Gradient Boosting	Data-driven relationships	Flexible, nonlinear modeling	Reduced interpretability	Emergence of deep learning
Deep Learning	RNN, LSTM	Learn temporal dependencies	Captures long-range patterns	Data-intensive and opaque	Development of hybrid models
Hybrid Models	ARIMA + Neural Networks	Separation of linear and nonlinear effects	Balanced structure and flexibility	Increased complexity	Integration of sentiment signals

Figure 2.1: Comparison of major cryptocurrency forecasting paradigms.

and illustrates how limitations in earlier approaches motivated the transition toward more flexible, hybrid, and sentiment-aware modeling frameworks.

A major shift in the evolution of cryptocurrency forecasting came with the introduction of deep learning architectures. Recurrent Neural Networks (RNNs) and especially Long Short-Term Memory (LSTM) networks introduced new capabilities for modeling sequential data that exhibits long-range dependencies. LSTMs address the vanishing gradient problem, which limits the ability of standard recurrent networks to learn long-term relationships, and retain information across extended sequences [8]. This allows them to capture the irregular, nonlinear, and highly volatile characteristics of cryptocurrency markets. Empirical

studies consistently show that LSTM-based models outperform traditional statistical and ensemble approaches when price dynamics exhibit complex temporal dependencies or sustained behavioral cycles [4, 9]. Variants such as Gated Recurrent Units, bidirectional networks, and attention-based models further extend these capabilities by capturing both short-term and long-term temporal relationships.

Recent work has increasingly focused on transformer architectures. Originally developed for natural language processing, transformers use self-attention mechanisms that identify dependencies across long sequences without relying on sequential computation. This makes them especially powerful in markets where long-range behavioral patterns, investor attention cycles, and regulatory or macroeconomic developments play important roles [11]. Their introduction signals a shift toward models that evaluate broader contextual information rather than focusing narrowly on recent price movements. Transformers also support multi-feature and multi-modal inputs, aligning well with the need to integrate price, volume, sentiment, and other external signals [11].

Despite their predictive power, deep learning models introduce challenges related to transparency, interpretability, and stability. The internal reasoning of these models is often difficult to trace, which can limit trust in their predictions. In financial contexts, this opacity is both a practical and ethical concern. Analysts and institutions require clear explanations of how inputs influence outputs in order to avoid reliance on unstable or spurious correlations. Models that perform well historically may fail under real-world stress if their predictions are driven by patterns that appear only under narrow conditions [11]. These concerns have shaped recent discussions surrounding responsible AI development in financial forecasting.

The evolution of cryptocurrency forecasting also reflects a growing recognition that prices are influenced by behavioral and psychological forces. Recent research increasingly incorporates sentiment features derived from social media, search trends, and news tone [7, 5]. These studies demonstrate that investor mood and public attention often precede significant price movements. Incorporating sentiment-based features into forecasting models

has therefore become an important direction in modern research, particularly for assets such as Bitcoin that respond strongly to social dynamics [7, 1].

Taken together, this evolution shows that cryptocurrency forecasting has moved from transparent but structurally limited statistical models toward more flexible machine learning and deep learning approaches that can better accommodate nonlinear and unstable market behavior. This progression provides the foundation for the benchmark design used in this chapter, which compares Naïve persistence, ARIMA, and LSTM as three representative price-only forecasting models of increasing complexity. The next section introduces the Bitcoin dataset, experimental design, and evaluation framework used to assess these models under consistent conditions.

2.2 PRICE-ONLY FORECASTING

Before incorporating sentiment-based or behavioral signals, it is necessary to establish a realistic price-only benchmark on the real Bitcoin dataset used in the *Momentum AI* framework. The purpose of this analysis is to determine how much predictive structure can be extracted from historical prices alone and to create a transparent baseline against which later sentiment-aware models can be evaluated.

This section evaluates three price-only forecasting approaches on the real Bitcoin dataset used later in the full *Momentum AI* pipeline: a naïve persistence baseline, a classical ARIMA model, and a simple univariate Long Short-Term Memory (LSTM) network. The objective is to establish realistic benchmark performance under a shared and consistent evaluation window, and to clarify what predictive power can be extracted from historical prices alone.

Bitcoin presents a uniquely challenging forecasting environment. The market exhibits strong nonstationarity, volatility clustering, momentum bursts, abrupt regime shifts, and exogenous shock responses driven by macroeconomic events, regulatory developments, and

collective behavioral dynamics. These structural properties strain the assumptions of classical time series models and place fundamental limits on price-only prediction.

Figure 2.2 displays the daily Bitcoin closing price over the past five years. Extended bull and bear phases, sharp accelerations, and clustered volatility are visually apparent. These features motivate the use of multiple baseline models and provide context for the performance behavior observed throughout this chapter.



Figure 2.2: Daily Bitcoin closing price over the past five years.

All trained models (ARIMA and LSTM) in this chapter are evaluated using a chronological 80/20 train-test split on data spanning January 2023 through January 2025. The resulting shared test window covers August 7, 2024 through December 31, 2024. This consistent evaluation period ensures comparability across models and isolates differences in modeling

assumptions rather than differences in sample selection. The models examined span increasing levels of complexity. The naïve model serves as a persistence benchmark. The ARIMA model introduces a wider historical context. The LSTM adds nonlinear sequence modeling capacity. All models are restricted to historical prices only. No external signals are incorporated. This design isolates the informational content of price history and clarifies its predictive limitations.

Model performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Squared Error (MSE). Let y_i denote the actual observed value, let $\hat{y}_i$ denote the corresponding predicted value, and let n denote the number of observations in the evaluation set. These metrics are defined as

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|,$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

and

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}.$$

MAE measures the average absolute magnitude of prediction error in the original price scale. MSE measures the average squared prediction error and is used directly in the ARIMA model selection procedure. RMSE is the square root of MSE, which places the error back on the original price scale while penalizing larger deviations more heavily.

2.2.1 NAÏVE PERSISTENCE BASELINE

The naïve forecasting model is the simplest possible time series predictor. It assumes that the best estimate of tomorrow's price is today's observed price, formally defined as

$$\hat{y}_{t+1} = y_t. \tag{2.1}$$

Despite its simplicity, this baseline is widely used in financial time series analysis because many asset prices exhibit short-term persistence and random-walk-like behavior. As a result, a naïve forecast can be surprisingly difficult to outperform, particularly at short horizons. Any forecasting model that fails to exceed this baseline provides limited practical value. Using the shared test window (August 7, 2024–December 31, 2024), the naïve model achieves an MAE of 1433.10 USD, an RMSE of 2017.70 USD, and an MSE of 4,071,107.29 USD². Figure 2.3 shows the naïve forecast over the evaluation period.

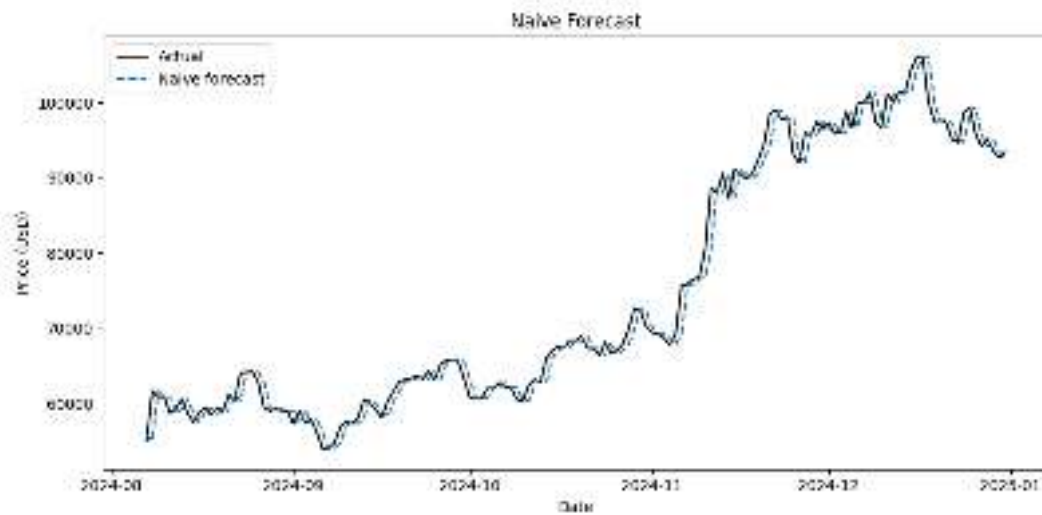


Figure 2.3: Naïve persistence forecast over the shared Bitcoin test window.

The model tracks directional movement closely because it effectively mirrors yesterday’s price. During sustained momentum phases within the test window, this persistence mechanism proves highly competitive. However, the naïve model cannot anticipate regime shifts or volatility shocks; it merely follows realized movement. This makes it both a strong benchmark and a useful structural diagnostic. If a more sophisticated model cannot consistently outperform the naïve baseline, the limitation is often informational rather than architectural.

The relatively low MAE and RMSE indicate that, within this evaluation window, simple persistence captures a large share of the short-horizon movement in Bitcoin prices. This happens because the test period contains sustained directional momentum, and the naïve

model immediately inherits that movement from the last observed price. Its success therefore reflects the structure of the window more than any deep modeling intelligence. The model performs well when the market continues moving in the same direction, yet it remains structurally incapable of anticipating reversals before they occur.

2.2.2 ARIMA FORECASTING

The second price-only model evaluated in this chapter is the Autoregressive Integrated Moving Average (ARIMA) model. Unlike the naïve baseline, ARIMA explicitly models linear temporal dependence and short-term autocorrelation in the data. An $\text{ARIMA}(p, d, q)$ model combines three components:

- an autoregressive component of order p , which captures dependence on past values,
- a differencing component of order d , which is used to reduce nonstationarity,
- a moving average component of order q , which models past forecast errors.

A standard representation of an ARIMA model is

$$\phi(L)(1 - L)^d y_t = c + \theta(L)\epsilon_t, \quad (2.2)$$

where y_t is the observed series at time t , L is the lag operator, c is a constant term, and ϵ_t is a white-noise error term. The lag operator shifts the series backward in time, so that $Ly_t = y_{t-1}$, $L^2 y_t = y_{t-2}$, and so on. Writing the model in terms of L provides a compact way to describe dependence on multiple past observations.

At a high level, ARIMA attempts to answer three questions. First, how many past observations should influence the present? Second, how much differencing is needed to stabilize the series? Third, how much of the recent error structure should be used to adjust the forecast? The parameters (p, d, q) correspond directly to these questions.

The autoregressive component, represented by the polynomial $\phi(L)$, captures dependence on past values of the series. In expanded form, this part can be written as

$$\phi(L) = 1 - \phi_1 L - \phi_2 L^2 - \cdots - \phi_p L^p, \quad (2.3)$$

so the coefficients $\phi_1, \phi_2, \dots, \phi_p$ determine how strongly past prices influence the present. Intuitively, a small p means the model only looks back a short distance, while a larger p means the model relies on a longer linear memory. Very small values of p may miss useful short-run persistence, whereas overly large values may cause the model to chase noise and become unnecessarily complex.

The integrated component, represented by $(1 - L)^d$, performs differencing. Differencing transforms the original series into a new series based on changes rather than levels. When $d = 1$, the model uses first differences,

$$(1 - L)y_t = y_t - y_{t-1}, \quad (2.4)$$

which means the model is no longer forecasting the raw price directly, but rather the day-to-day change in price. When $d = 2$, differencing is applied again, producing second differences. The purpose of differencing is to reduce nonstationarity. Many financial price series exhibit trend, drift, or changing mean levels over time, which violate the assumptions of linear time-series models. A low value of d may leave too much nonstationarity in the series, while a high value may over-difference the data and remove useful signal. In practice, values of d are usually small, often 0, 1, or 2.

The moving-average component, represented by $\theta(L)$, captures dependence on past forecast errors rather than past observed values. It can be written as

$$\theta(L) = 1 + \theta_1 L + \theta_2 L^2 + \cdots + \theta_q L^q. \quad (2.5)$$

This part allows the model to adjust current forecasts based on how wrong it was in

previous periods. If recent residuals show systematic overprediction or underprediction, the moving-average terms can partially correct for that pattern. Intuitively, a larger q gives the model more flexibility in responding to recent mistakes, but excessively large values can again increase instability and encourage overfitting.

Taken together, these three components make ARIMA a flexible but still interpretable linear forecasting framework. The autoregressive terms model persistence, the differencing term addresses trend and nonstationarity, and the moving-average terms account for short-run error structure. This is one reason ARIMA remains a common benchmark in forecasting research: it provides a transparent way to test whether a time series contains exploitable local linear structure.

ARIMA order selection was performed through a grid search over candidate (p, d, q) combinations, with the final specification chosen according to out-of-sample mean squared error under the shared evaluation design.

The best-performing specification on the shared test window was ARIMA(3, 1, 2). This means the final model uses three autoregressive lags, first differencing, and two moving-average terms. Intuitively, this specification assumes that short-run Bitcoin price changes retain some dependence on the previous three observations, require one level of differencing to reduce nonstationarity, and benefit from correcting forecasts using the most recent two error terms.

On the same August 2024 through December 2024 test window, ARIMA(3, 1, 2) achieves an MAE of 18538.71 USD, an RMSE of 24727.77 USD, and an MSE of 611,462,604.17 USD². Figure 2.4 shows the ARIMA forecast.

While ARIMA captures short-term autocorrelation, its forecasts remain overly smooth and slow to adjust to changing market conditions. During nonlinear acceleration phases in late 2024, the model responds too gradually, leading to substantially larger errors relative to the naïve baseline. In practical terms, the model underperforms because the dominant movements in the test window reflect structural shifts, nonlinear momentum, and volatility

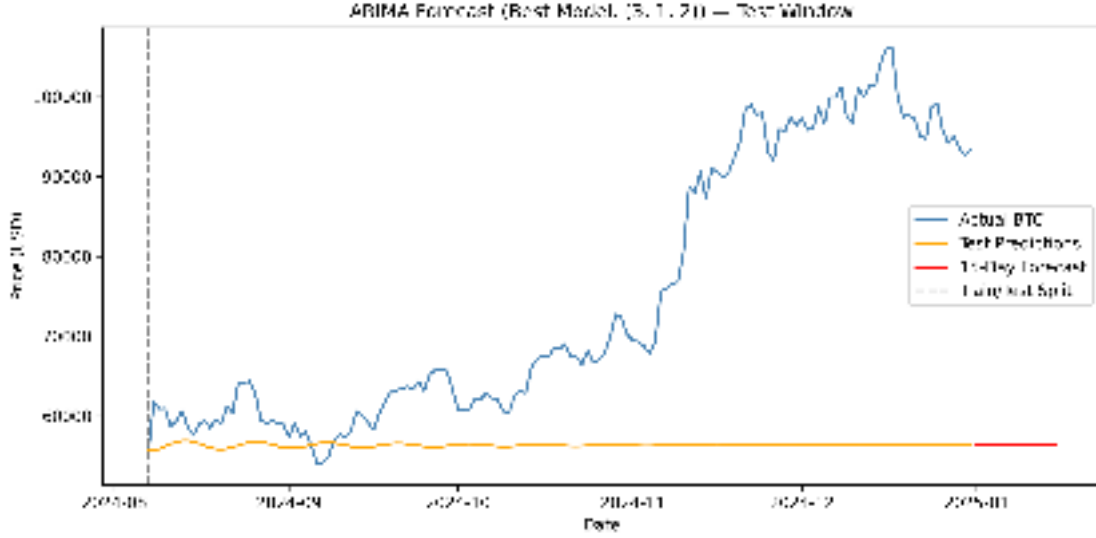


Figure 2.4: ARIMA(3, 1, 2) forecast over the shared Bitcoin test window. The dashed gray vertical line marks the chronological boundary between the training data and the held-out test window.

expansion rather than stable local linearity. This outcome highlights the difficulty of modeling regime-dependent volatility with stationary linear processes.

ARIMA’s inclusion remains analytically important despite this weak final performance. A benchmark is useful not only when it performs well, but also when it clarifies what kind of structure the data does or does not contain. In this case, ARIMA serves as a test of whether Bitcoin’s short-run dynamics can be described adequately through stable local linearity. Its underperformance therefore provides evidence that the dominant behavior in this window is not primarily linear, but instead shaped by rapidly changing and regime-dependent market conditions.

2.2.3 PRICE-ONLY LSTM FORECASTING

To evaluate whether nonlinear temporal modeling alone improves price-only forecasting, this section introduces a univariate LSTM trained on the same dataset and evaluated on the identical shared test window.

LSTMs are a class of recurrent neural networks specifically designed to model sequential

data with long-range temporal dependencies. Unlike linear models, LSTMs can learn nonlinear relationships between past observations and future outcomes. Importantly, in this experiment the LSTM is provided only with historical price information. No sentiment, volume, or external signals are included. This isolates the effect of nonlinear sequence modeling and allows a clean comparison with the price-only baselines introduced earlier.

Prices are normalized to $[0, 1]$ using a MinMaxScaler fitted exclusively on the training data:

$$\tilde{p}_t = \frac{p_t - p_{\min}}{p_{\max} - p_{\min}}. \quad (2.6)$$

A sliding window of $N = 60$ days is used to construct input sequences:

$$X_t = [\tilde{p}_{t-60}, \dots, \tilde{p}_{t-1}], \quad y_t = \tilde{p}_t. \quad (2.7)$$

The lookback window determines how much historical information the model sees before making each prediction. In this case, a 60-day lookback means that the LSTM receives the previous sixty days of normalized closing prices and then predicts the next day. The choice of lookback matters because it shapes the balance between short-term responsiveness and broader sequential memory.

The 60-day window was selected as a practical compromise between memory length and model simplicity. A shorter window, such as 7, 14, or 30 days, would make the model more responsive to very recent movements, but it would also reduce the amount of historical context available for learning broader sequential structure. In a market such as Bitcoin, where price swings can unfold across several weeks, an overly short window risks emphasizing local noise at the expense of medium-term momentum and regime development. By contrast, a much longer window could provide more context, but it would also increase dimensionality, slow training, and potentially make the model less responsive to recent changes. The 60-day choice therefore reflects a balance: it is long enough to capture medium-term directional

behavior, and still short enough to keep the model tractable and focused on relatively recent price dynamics.

Because of this 60-day lookback, the LSTM shares the same overall train–test split as the naïve and ARIMA models, but its effective prediction period begins later. Within the broader August 7, 2024 through December 31, 2024 test window, the LSTM evaluation begins on October 6, 2024. This difference is methodological rather than conceptual: the split is shared, but sequence construction requires an initial lookback period before the first out-of-sample prediction can be generated.

The LSTM architecture consists of:

- one LSTM layer with 64 hidden units,
- dropout with rate 0.2,
- one dense output layer.

The model is trained using the Adam optimizer and mean squared error loss. The Adam optimizer is widely used in deep learning because it adapts the learning rate during training and tends to converge efficiently on noisy optimization problems. This is especially useful in financial time series settings, where the loss surface can be unstable and the data itself can be highly variable. In this thesis, Adam is used because it provides a strong and standard optimization baseline for comparing the price-only and sentiment-aware LSTM settings under the same training logic.

Early stopping is employed to prevent overfitting, and temporal ordering is preserved throughout training. The architecture is intentionally simple. The purpose is not to optimize a highly tuned deep learning model, but to test whether nonlinear sequence modeling alone can extract predictive value from price history beyond what is achieved by the simpler baselines.

On its evaluation window, the price-only LSTM achieves an MAE of 2824.59 USD, an

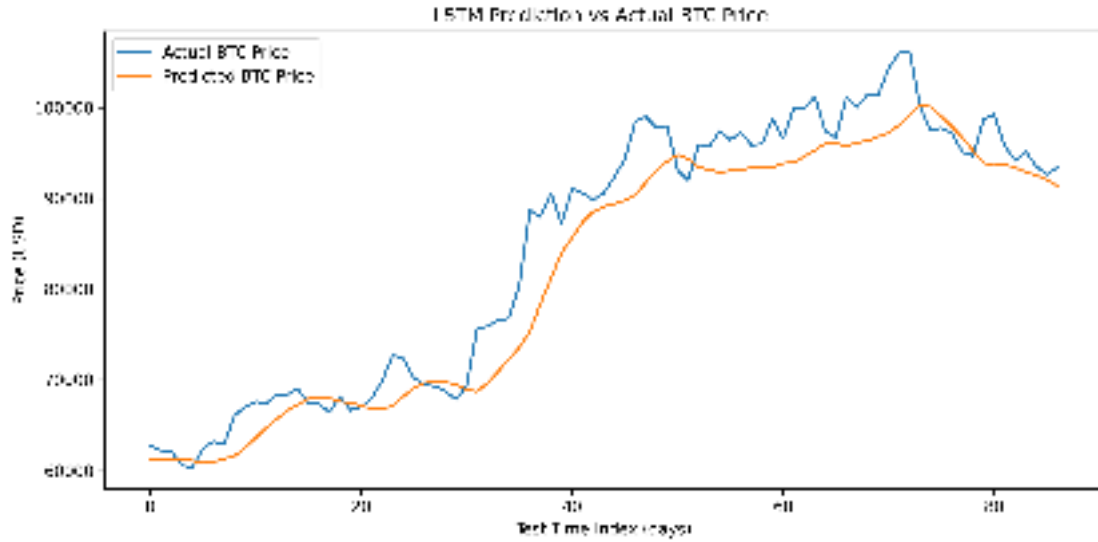


Figure 2.5: Price-only LSTM predictions versus true Bitcoin prices on the shared test window. RMSE of 3720.86 USD, and an MSE of 13,844,799.22 USD². Figure 2.5 shows the model’s predictions against the true Bitcoin prices on the shared test window.

Because the model uses a 60-day lookback window, its effective evaluation begins later within the broader August–December 2024 test period. The LSTM captures broad directional movement but produces smoother forecasts that lag sharp accelerations. This smoothing behavior arises from optimization under squared-error loss, which penalizes large deviations and encourages regression toward the conditional mean. The model is therefore able to follow the overall trend more effectively than ARIMA, but it still underreacts to abrupt price movements.

Compared with ARIMA, the LSTM achieves dramatically lower error, which suggests that nonlinear sequence modeling captures more of Bitcoin’s short-run structure than a linear autoregressive process. However, the LSTM still does not beat the naïve benchmark. This is analytically important. It shows that when the dominant signal in the market is local persistence, even a flexible nonlinear model may offer limited advantage over simply carrying forward the most recent observation. In other words, additional modeling flexibility helps, yet it does not automatically overcome the strength of short-run momentum. A shorter-window

LSTM might plausibly react faster to abrupt local changes, but it would likely sacrifice some ability to learn the broader trend structure visible over multiweek periods. Because this thesis prioritizes a clean benchmark comparison rather than exhaustive hyperparameter tuning, the 60-day window is treated as a reasonable and interpretable baseline rather than as a claim of global optimality.

2.3 COMPARATIVE RESULTS ACROSS MODELS (RQ1)

Table 2.1: Comparison of price-only forecasting performance on the shared Bitcoin evaluation window. Lower values indicate better predictive performance.

Model	MAE (USD)	RMSE (USD)	MSE (USD ²)
Naïve Persistence	1,433.10	2,017.70	4,071,107.29
ARIMA(3, 1, 2)	18,538.71	24,727.77	611,462,604.17
Price-only LSTM	2,824.59	3,720.86	13,844,799.22

Table 2.1 makes the relative performance differences clear. The Naïve persistence model achieves the lowest error across all three metrics, which indicates that short-run local continuity dominates this particular evaluation window. In practical terms, yesterday’s price is often already a strong predictor of today’s price when the market is moving in sustained directional phases.

The ARIMA model performs substantially worse than both alternatives. Its high MAE shows that its average forecast misses are large in absolute terms, while its RMSE and MSE indicate especially poor performance during larger deviations. This matters because RMSE and MSE penalize large errors more heavily than MAE. Under the same evaluation conditions, ARIMA(3, 1, 2) does not capture the observed test-window dynamics as effectively as either the Naïve baseline or the price-only LSTM. As discussed earlier, ARIMA(3, 1, 2) was retained because multiple candidate (p, d, q) specifications were tested and this configuration produced

the strongest out-of-sample performance among the ARIMA candidates considered, even though its final test performance remained weak relative to the other model classes.

The price-only LSTM performs far better than ARIMA but still worse than the Naïve benchmark. Its lower MAE, RMSE, and MSE relative to ARIMA show that nonlinear sequence modeling captures more of Bitcoin’s temporal structure than a linear autoregressive framework. At the same time, the LSTM’s inability to beat Naïve persistence indicates that nonlinear flexibility alone is not enough when the dominant short-run signal is simple momentum. In this benchmark window, simply carrying forward the most recent observed price produces smaller one-step-ahead errors than the smoother LSTM forecasts.

This pattern does not mean that the Naïve approach is universally best. Rather, it shows that in this specific period, Bitcoin’s short-run behavior is governed more by local continuation than by stable linear structure or by nonlinear sequence learning that can anticipate turning points in advance.

More broadly, the comparative results in this chapter highlight a central limitation of price-only modeling. Increasing model complexity does not automatically compensate for restricted informational input. During strong momentum regimes, simple persistence remains highly competitive, which helps explain the strong performance of the Naïve baseline in the shared evaluation window. By contrast, ARIMA struggles under structural breaks, volatility expansion, and rapid nonlinear acceleration. The LSTM captures broader directional structure than ARIMA, but it still tends to smooth the series rather than anticipate abrupt regime changes.

Taken together, these findings suggest that forecasting limitations in cryptocurrency markets are primarily informational rather than computational. When price dynamics are shaped by external events, narrative shifts, and behavioral reactions, models restricted to historical prices remain limited in their ability to anticipate abrupt transitions. The Naïve baseline performs well when short-run momentum dominates, ARIMA underperforms when

local linearity breaks down, and the LSTM, although more flexible, remains constrained by the same informational boundary as the other price-only models.

RQ1 Answer: RQ1 asks, “Which forecasting models perform best for short-term cryptocurrency price prediction when evaluated under identical experimental conditions?” Under the shared price-only Bitcoin benchmark, the Naïve persistence model performs best, outperforming both ARIMA(3, 1, 2) and the price-only LSTM across MAE, RMSE, and MSE. This indicates that short-horizon Bitcoin forecasting is highly regime dependent and that greater model complexity alone does not overcome the informational limits of historical price data.

These limitations become especially important once sentiment enters the forecasting problem. The Naïve model cannot incorporate external signals at all, while ARIMA remains poorly suited to the nonlinear, asymmetric, and regime-dependent ways in which sentiment interacts with price. Even when exogenous variables are introduced, linear models struggle to represent how emotional responses accumulate, decay, or intensify under different market conditions. This leaves more flexible sequence models as the most natural setting for sentiment-aware forecasting, but only if sentiment can first be transformed into a structured and interpretable signal.

Rather than continuing to increase architectural complexity within a price-only setting, the analysis now turns toward expanding the informational input space. The next chapter therefore develops the behavioral and sentiment framework underlying cryptocurrency market dynamics, examines the Alternative.me Fear and Greed Index, and introduces the transparent Momentum Fear and Greed Index used in the later forecasting analysis.

CHAPTER 3

SENTIMENT IN CRYPTOCURRENCY MARKETS

Cryptocurrency markets differ fundamentally from traditional financial systems in their sensitivity to emotion, online discourse, and rapid shifts in collective attention. Unlike equities or commodities, which are often anchored by earnings, balance sheets, and macroeconomic indicators, cryptocurrencies operate in an environment dominated by speculation, narrative formation, and behavioral feedback loops. Trading occurs continuously across global exchanges, information propagates almost instantaneously through online platforms, and collective reactions often materialize before any formal economic interpretation can take place. As a result, price movements are frequently driven by social, emotional, and psychological forces rather than by intrinsic valuation.

This structural distinction has important implications for forecasting. In traditional markets, prices tend to reflect slowly evolving fundamentals and institutional decision-making processes. In contrast, cryptocurrency prices often respond to changes in public mood, online narratives, and perceived momentum. Sudden shifts in sentiment can produce rapid price movements that appear disconnected from underlying technological or economic developments. A forecasting framework that relies exclusively on historical price information therefore captures only a partial representation of market dynamics. Such models may describe persistence or short-term trends, but they struggle to anticipate abrupt transitions driven by behavioral factors.

These characteristics make cryptocurrency forecasting a behavioral as well as a statistical

problem. Price-only models capture part of the market's temporal structure, but they often miss the emotional and narrative forces that shape short-horizon movements. This chapter therefore develops the sentiment and behavioral framework used in the thesis, beginning with the role of sentiment in cryptocurrency markets, then examining the Alternative.me Fear and Greed Index (AFGI), and finally introducing the transparent Momentum Fear and Greed Index (MFGI).

3.1 ROLE OF SENTIMENT

Traditional economic theory assumes that investors behave rationally and process information efficiently. Behavioral finance challenges this assumption by demonstrating that emotions, cognitive shortcuts, and social influences heavily shape real-world decision-making. These forces are especially visible in cryptocurrency markets, where enthusiasm, fear, and online social dynamics spread rapidly and influence price movements at a global scale. The continuous nature of crypto trading, combined with its highly engaged online communities, allows sentiment-driven signals to propagate quickly and intensify collective market reactions.

Social media platforms play a central role in shaping cryptocurrency sentiment. Twitter, Reddit, Telegram, Discord, and YouTube act as gathering spaces where information, rumors, and emotional reactions circulate in real time. Positive narratives such as anticipated technological upgrades, endorsements by influential figures, or speculative excitement often trigger rapid periods of buying activity fueled by fear of missing out. Negative narratives, such as security breaches or regulatory threats, can just as quickly spark panic selling and widespread retreat from risk. The rapid feedback loop between sentiment and price amplifies volatility, creating conditions where market psychology becomes as important as market fundamentals.

Early efforts to quantify sentiment relied on lexicon-based methods that classify words as positive or negative based on dictionary rules. These methods provided useful initial insights

but struggled with sarcasm, slang, evolving crypto terminology, and culturally specific online communication styles. Modern sentiment research has shifted toward more advanced natural language processing methods, particularly transformer-based language models that interpret sentiment in context. These models generate richer sentiment embeddings that capture tone, linguistic nuance, and thematic emphasis. Combined with sequence models such as Long Short-Term Memory networks, they help reveal how sentiment evolves over time and how it correlates with price movements.

Sentiment plays an unusually central role in cryptocurrency markets because these markets lack many of the stabilizing mechanisms present in traditional finance. Conventional assets benefit from institutional participation, regulatory oversight, and valuation anchors that tend to moderate excessive volatility and dampen speculative extremes. Cryptocurrencies are shaped primarily by retail participation, speculative capital, and persistent online engagement. In this environment, public mood, collective narratives, and perceived momentum often act as primary drivers of price dynamics, particularly during periods of rapid appreciation or sudden decline.

This makes cryptocurrency markets especially vulnerable to narrative contagion. A rumor, tweet, exchange event, regulatory headline, or influential endorsement can spread quickly through online communities and reshape market expectations before any deeper economic interpretation takes place. In this sense, price movement in crypto is often socially amplified. Traders do not respond only to fundamentals; they respond to what they believe other traders are feeling and about to do. This reflexive structure makes behavioral context especially important for understanding short-horizon movements.

Another reason sentiment matters so strongly in crypto is the speed with which market narratives form and collapse. In traditional financial environments, interpretation often moves through institutional channels such as earnings reports, analyst revisions, and macroeconomic releases. In cryptocurrency markets, narratives can emerge from loosely connected online communities, influencer commentary, exchange announcements, or rapidly circulated rumors.

This means that shifts in emotion often appear before a clear economic explanation is available. A forecasting system that ignores this environment risks treating emotionally driven market swings as random noise, even when they follow recognizable patterns of public attention and collective reaction.

Empirical research consistently shows that cryptocurrency returns exhibit sharp regime shifts, volatility clustering, and heavy-tailed distributions, all of which reflect the influence of behavioral forces rather than stable economic fundamentals [10]. These features suggest that price dynamics are governed by past values, changing expectations, and emotional responses. Market participants frequently react to signals such as online discussion intensity, sentiment polarity, and narrative framing, even when no concrete economic information is introduced.

Search interest provides another valuable lens into public attention and collective emotion. Google Trends data reveals how frequently users search for cryptocurrency-related queries, offering a proxy for shifts in public concern, curiosity, or fear. Figure 3.1 illustrates this behavior using global search interest for the query *Bitcoin crash* between 2021 and 2022. Peaks in search activity align with periods of market stress, indicating heightened public attention and emotional response during downturns.

Search behavior is especially useful because it captures a wider audience than market commentary alone. Social-media discussions often reflect the views of highly active crypto participants, whereas search spikes can indicate broader public concern, confusion, or urgency. When users begin searching phrases such as *Bitcoin crash*, *crypto collapse*, or *should I sell Bitcoin*, they reveal a rise in anxiety and information-seeking behavior that may correspond to a worsening emotional regime. Search interest therefore complements text sentiment by offering another way to observe how collective concern moves through the market.

Studies integrating sentiment indicators into forecasting frameworks consistently find that measures of public mood and attention contain useful information beyond historical prices alone, particularly over short-term horizons where emotional responses dominate rational reassessment [7, 5]. In cryptocurrency markets, sentiment can also be anticipatory. Shifts in

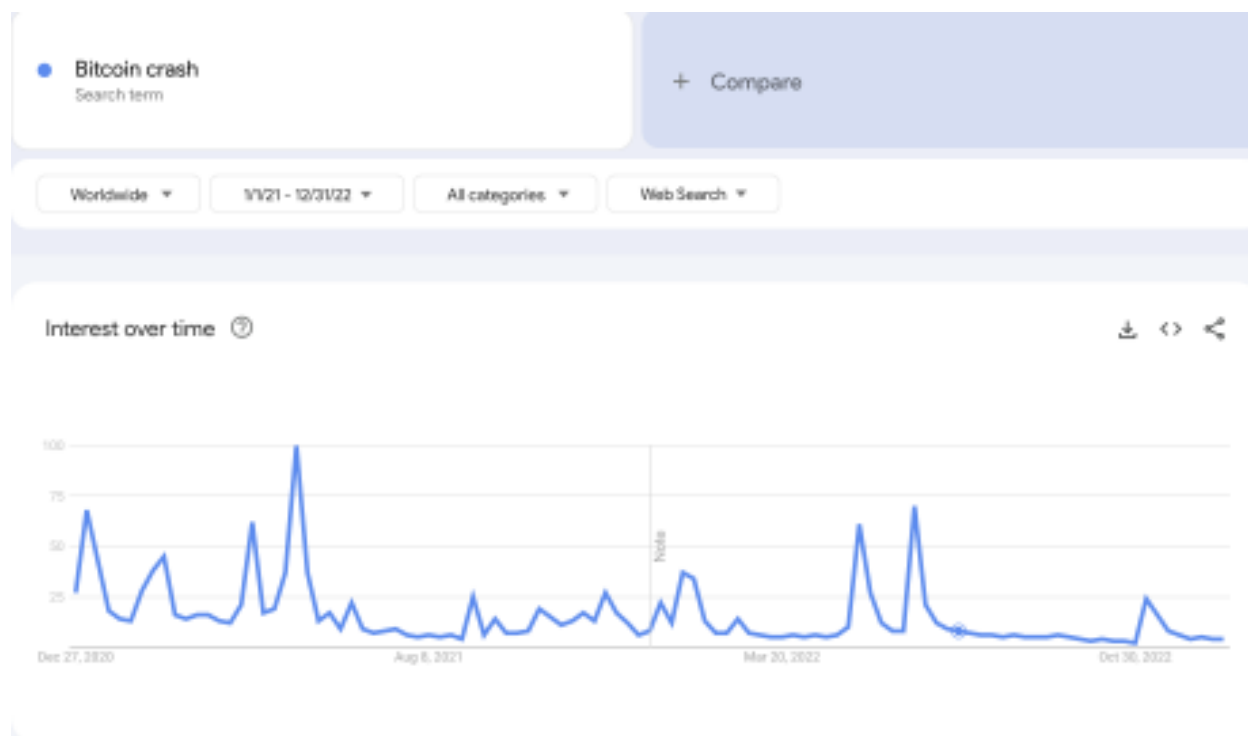


Figure 3.1: Google Trends search interest for the query “Bitcoin crash” from 2021 to 2022. Peaks in search activity reflect heightened public attention and fear during periods of market stress, illustrating the role of behavioral signals in cryptocurrency markets.

online tone, discussion volume, or search behavior frequently precede major price movements rather than simply following them.

Taken together, these observations show that cryptocurrency markets are influenced by historical price behavior, shifting narratives, collective attention, and investor emotion. Sentiment and attention-based indicators therefore provide a meaningful behavioral layer that complements price data and helps explain why digital asset markets often experience abrupt and amplified movements. This behavioral backdrop motivates the use of structured sentiment measures that summarize market mood in a form suitable for interpretation and forecasting.

3.2 THE ALTERNATIVE.ME FEAR AND GREED INDEX (AFGI)

Among existing sentiment indicators, the Alternative.me Fear and Greed Index (AFGI) is one of the most widely referenced public measures of cryptocurrency market mood. The index is reported on a scale from 0 to 100, where lower values indicate stronger market fear and higher values indicate stronger market greed. It presents a simple daily score intended to summarize whether the market is operating in a state of fear or greed, making it attractive to traders, commentators, and market observers seeking a quick behavioral snapshot. Its appeal lies in its accessibility: rather than requiring users to interpret multiple market variables separately, the index offers a compact representation of sentiment conditions in a single number and visual scale.

In practice, the AFGI functions as a shorthand for the emotional tone of the market. It is commonly cited in media coverage, investor discussions, and informal market analysis as an indicator of whether conditions appear excessively pessimistic or overly euphoric. This kind of index is especially attractive in cryptocurrency settings, where rapid sentiment shifts, speculative surges, and fear-driven selloffs often shape short-horizon price dynamics. Figure 3.2 shows a widely referenced AFGI interface. Its visual design makes the index intuitive and easy to interpret, which helps explain its popularity in practice.

The popularity of the AFGI also helps clarify why fear-and-greed style indicators matter in the first place. Traders and commentators often want a quick summary of whether the market appears stretched, anxious, overheated, or complacent. In that sense, the AFGI succeeds as a communication tool: it gives market participants a compact way to discuss sentiment without separately interpreting multiple variables.

Its popularity also reveals an important point about market behavior itself. Traders clearly value broad sentiment summaries. AFGI-style indicators continue to circulate because they satisfy a real interpretive need. Participants want a quick sense of whether the market is becoming euphoric, unstable, fearful, or overheated. That demand is meaningful for this



Figure 3.2: Example of a widely referenced Fear and Greed Index interface. While such visualizations provide intuitive summaries of market emotion, their underlying data sources, feature definitions, and weighting schemes are often undisclosed, limiting their scientific interpretability and transparency.

thesis because it suggests that regime-style emotional indicators are useful to people already operating in the market, even before asking whether they improve formal forecasting models.

Despite its practical appeal, the AFGI presents several limitations from a research perspective. The first is methodological opacity. Commercial implementations often rely on proprietary formulas that do not fully disclose how component signals are selected, normalized, or weighted. This makes it difficult to determine whether reported index values reflect meaningful sentiment dynamics or arbitrary design choices. A second limitation concerns feature transparency. Inputs such as volatility, social activity, or search behavior may be referenced, but the exact preprocessing rules and aggregation procedures are not always made fully visible. A third limitation concerns historical interpretability. Without complete documentation of the raw inputs and transformation rules, researchers cannot confidently trace why the index moved the way it did at particular points in time.

These limitations matter because they affect more than technical neatness. In a research setting, opacity makes it difficult to audit claims, compare methods fairly, or determine whether the signal is genuinely behaviorally informative rather than merely cosmetically

persuasive. A fear-and-greed score may look compelling on a dashboard while still remaining analytically weak if the logic behind it cannot be clearly examined.

There is also a practical limitation tied to time-series analysis. When an index is produced through undisclosed transformations, researchers cannot easily determine whether changes in the series reflect actual market conditions or shifts in the underlying construction process. This becomes especially important when interpreting historical episodes. If an index dips sharply during a crisis or rises quickly during a euphoric rally, it matters whether that movement can be explained by visible components or only observed as a finished output. For rigorous forecasting work, interpretability requires more than a published number. It requires a visible chain from raw input to final signal.

The AFGI still has value. Its continued visibility suggests that it captures behaviorally meaningful aspects of cryptocurrency markets. At the same time, its opacity makes it difficult to use as a rigorous research instrument. This creates the need for a transparent alternative whose components, transformations, and weighting choices can be fully inspected. The next section introduces that alternative through the construction of the Momentum Fear and Greed Index (MFGI).

3.3 THE MOMENTUM FEAR AND GREED INDEX (MFGI)

The limitations of existing fear and greed measures create a clear need for a more transparent alternative designed specifically for research use. In cryptocurrency markets, a sentiment index is most useful when it provides an interpretable summary of the broader behavioral environment in which price movements occur. For that reason, the index developed in this thesis is designed as a fully documented framework whose components, transformations, and weighting choices remain visible throughout.

The resulting Momentum Fear and Greed Index (MFGI) captures broad market state rather than each short-run price movement directly. Its purpose is to describe whether

market conditions appear relatively fearful, neutral, or greed-dominated by combining three dimensions that are behaviorally meaningful in cryptocurrency settings: price momentum, realized volatility, and aggregated daily news sentiment. Together, these components provide a more interpretable representation of market context than price history alone, while avoiding the opacity that limits the usefulness of many commercial sentiment indicators.

The logic of the index is straightforward. Momentum captures directional continuation and speculative drive. Volatility captures uncertainty, stress, and instability. Sentiment captures narrative tone and the emotional environment surrounding the market. None of these dimensions alone is sufficient to represent fear or greed fully. Combined, they provide a more balanced picture of market state than any one component in isolation.

This component structure also reflects the broader argument of the thesis. Fear and greed in cryptocurrency markets do not emerge from a single channel. They are visible in how people talk, how prices trend, and how unstable the market becomes. A composite index is therefore more appropriate than a one-dimensional sentiment measure for representing behavioral market state. It allows emotion, directional force, and uncertainty to be interpreted together rather than in isolation.

3.3.1 SENTIMENT PREPROCESSING AND DISCRETIZATION

The sentiment dataset used in this thesis consists of Bitcoin-related news headlines, each associated with a sentiment score and a publication timestamp. While this format is useful for textual analysis, it is not directly suitable for time-series forecasting. The forecasting models developed later in the thesis require regularly sampled numerical inputs aligned with the daily Bitcoin closing price series. As a result, the raw headline-level sentiment data must be transformed into a daily, temporally aligned signal before it can be integrated into the broader forecasting framework.

A central challenge in working with news sentiment is temporal irregularity. Headlines are not published at fixed intervals, and the number of available headlines varies substantially

from day to day. Some dates are dominated by bursts of news coverage surrounding major market events, while other dates contain relatively few observations. If raw headline scores were used directly, days with heavier media coverage would exert disproportionate influence on the resulting signal. The preprocessing pipeline therefore aims to construct a sentiment representation in which each day contributes comparably to the final time series.

To address this issue, all headline-level sentiment observations were aggregated to the daily level. Each timestamp was converted to a calendar date, and all sentiment values published on the same day were grouped together. The daily sentiment score was then defined as the mean of all headline sentiment values observed on that date. This aggregation aligns the sentiment signal with the daily frequency of Bitcoin closing prices and also reduces noise by averaging across multiple headlines rather than relying on isolated extreme observations.

Figure 3.3 shows the resulting daily mean sentiment series over the sample period. The signal is visibly noisy and frequently fluctuates around a weakly positive or near-neutral baseline. Short-lived spikes and drops remain present, but the daily aggregation produces a more stable representation than the original irregular headline stream.

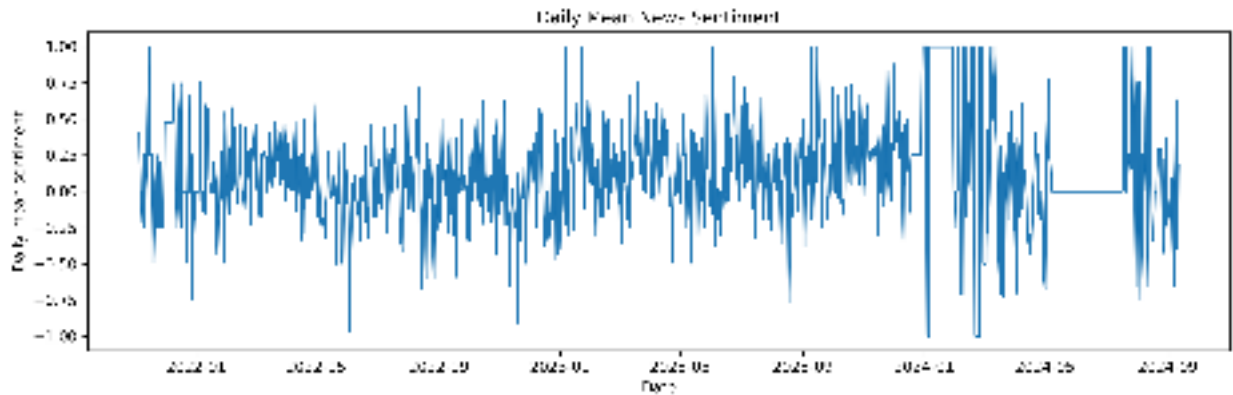


Figure 3.3: Daily mean Bitcoin news sentiment after aggregation from headline-level observations. The series remains noisy, but aggregation provides a temporally aligned daily signal suitable for downstream modeling.

The pattern in Figure 3.3 reinforces an important practical observation. Although sentiment varies meaningfully over time, most daily values remain relatively close to the

center of the scale, while more extreme positive and negative values occur less frequently. This suggests that the raw continuous sentiment signal contains useful behavioral information, but also substantial variability that must be interpreted carefully.

To better understand the distribution of this aggregated signal, Figure 3.4 presents a histogram of daily mean sentiment values. The distribution is concentrated near the middle, with thinner tails extending toward more extreme optimistic and pessimistic states.

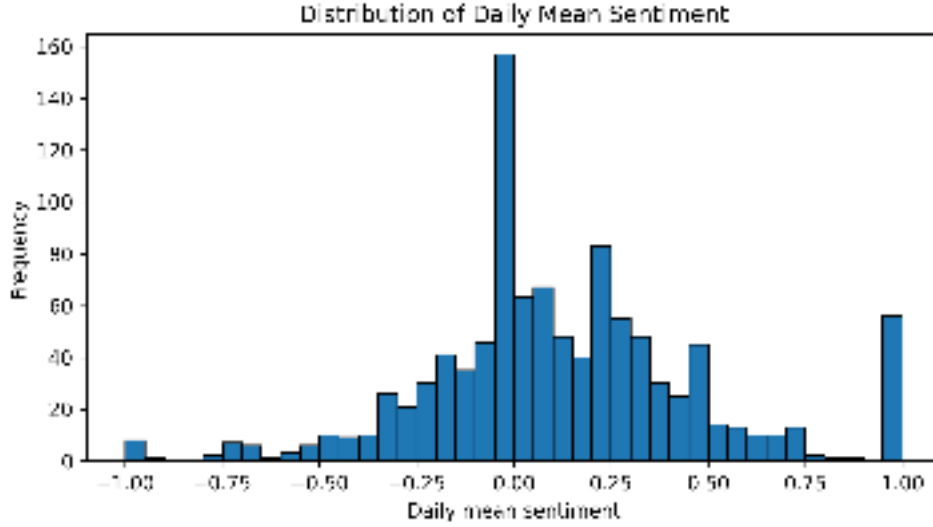


Figure 3.4: Distribution of daily mean sentiment values after daily aggregation. Most observations lie near the center of the distribution, indicating that strongly polarized sentiment days are comparatively less common.

This concentration near the center creates a modeling challenge. If the continuous sentiment signal is passed directly into a forecasting model, small numerical differences around neutrality may receive more attention than they deserve, while rare extreme values may dominate the signal in unstable ways. For this reason, the preprocessing pipeline also constructs a discretized version of sentiment that emphasizes broad emotional regimes rather than small day-to-day numerical fluctuations.

A quantile-based discretization strategy was employed. Specifically, the lower 33rd percentile of daily sentiment values was assigned to a *fear* state, the middle range was assigned to a *neutral* state, and the upper 33rd percentile was assigned to a *greed* state. These

classes were encoded numerically as -1 , 0 , and $+1$, respectively. Formally, the discretized sentiment signal is defined as

$$\text{sentiment_disc}_t = \begin{cases} -1, & \text{if } s_t \leq Q_{0.33}, \\ 0, & \text{if } Q_{0.33} < s_t < Q_{0.66}, \\ +1, & \text{if } s_t \geq Q_{0.66}, \end{cases}$$

where s_t denotes the daily mean sentiment score and $Q_{0.33}$ and $Q_{0.66}$ denote the empirical 33rd and 66th percentiles of the sentiment distribution.

This discretization serves several purposes. First, it converts a noisy continuous signal into a more interpretable regime-based representation. Second, it reduces the risk that forecasting models overfit minor sentiment fluctuations that are not behaviorally meaningful. Third, because the thresholds are based on quantiles rather than fixed hand-selected cutoffs, the procedure adapts naturally to the empirical distribution of the data.

Figure 3.5 shows the resulting class balance. The three classes are nearly evenly distributed, which confirms that the quantile-based discretization successfully avoids severe class imbalance while preserving a clear fear–neutral–greed interpretation.

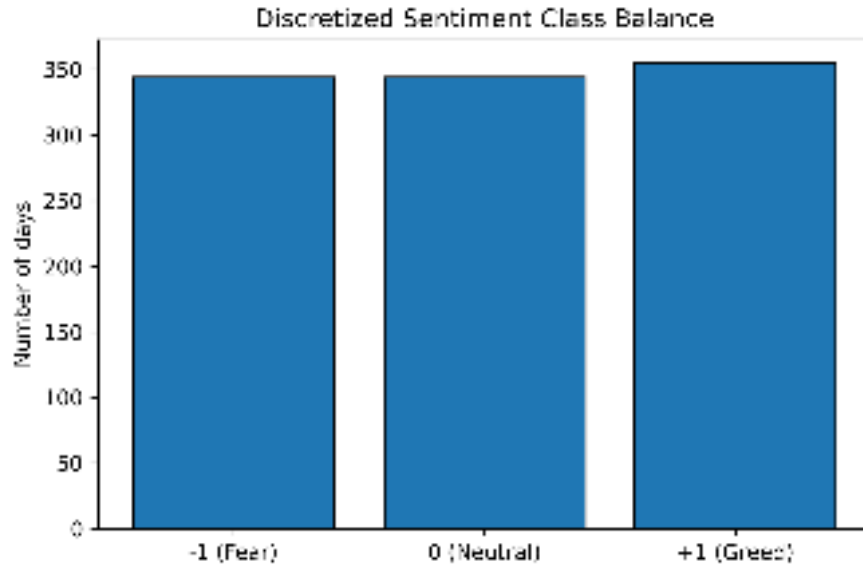


Figure 3.5: Class balance after quantile-based discretization of daily sentiment into fear (-1), neutral (0), and greed ($+1$) states. The classes are nearly evenly represented, supporting balanced regime-based modeling.

After discretization, the sentiment signal was aligned with the Bitcoin price series using daily calendar dates. This produced a joint daily dataset in which each price observation is associated with a corresponding sentiment value. Figure 3.6 visualizes the aligned relationship between Bitcoin closing prices and the daily mean sentiment signal which makes clear that sentiment and price interact in a complex way. In some periods, sentiment appears to shift before major price movements become fully visible. In other periods, sentiment reacts after prices have already moved. This asynchronous relationship is one reason sentiment is treated in this thesis as a behavioral context signal rather than a deterministic forecasting target.

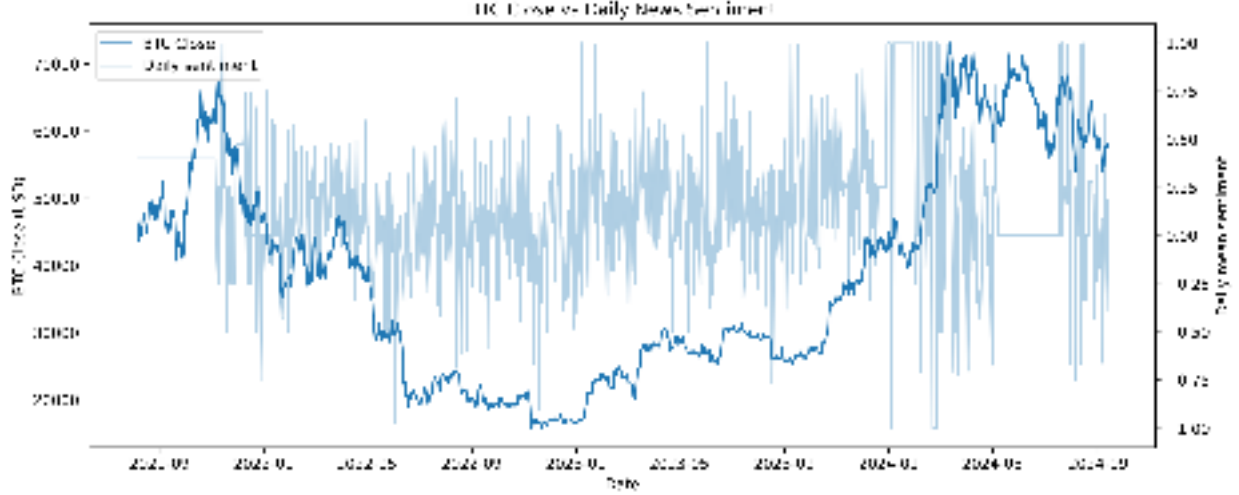


Figure 3.6: Bitcoin closing price and daily mean news sentiment aligned to the same daily calendar. The two series are related, but they do not move in lockstep, which motivates the use of sentiment as contextual information rather than as a direct proxy for price.

To preserve realistic temporal ordering, the sentiment features used in downstream models were lagged before being provided as inputs. In practice, this means that predictions at time t are based only on sentiment information that would have been available before time t . Three lagged versions of the signal were constructed so that later forecasting models, especially the sentiment-aware LSTM, could use only past sentiment information when generating predictions. A single lag did not provide much additional signal on its own, whereas three lagged values offered a short recent history of market mood without making the input structure unnecessarily wide or difficult to interpret. This made the one-day, two-day, and three-day lags a practical compromise between informational richness and model simplicity. Table 3.1 shows example rows from the resulting lagged feature table.

The lag structure matters for both interpretation and modeling. In the later sentiment-aware LSTM setting, these lagged sentiment variables are incorporated as additional input features alongside the historical price sequence. As a result, the model predicts future price movements using only information that would already have been observable at the forecast date. This allows the LSTM to learn whether recent sentiment conditions are strengthening,

Table 3.1: Example rows of lagged sentiment features. Both discretized and raw lagged sentiment values are shown, ensuring that only past information is available to later forecasting models.

Date	sentiment_raw	sentiment_disc	disc_lag1	disc_lag2	disc_lag3	sentiment_raw_lag1	sentiment_raw_lag2	sentiment_raw_lag3
2021-11-08	-0.2099	-1	-1.0	-1.0	1.0	0.0000	0.0000	0.3996
2021-11-09	0.2499	1	-1.0	-1.0	-1.0	-0.2099	0.0000	0.0000
2021-11-10	-0.2492	-1	1.0	-1.0	-1.0	0.2499	-0.2099	0.0000
2021-11-11	0.0000	-1	-1.0	1.0	-1.0	-0.2492	0.2499	-0.2099
2021-11-12	0.2499	1	-1.0	-1.0	1.0	0.0000	-0.2492	0.2499
2021-11-13	0.2499	1	1.0	-1.0	-1.0	0.2499	0.0000	-0.2492
2021-11-14	0.9996	1	1.0	1.0	-1.0	0.2499	0.2499	0.0000
2021-11-15	0.2499	1	1.0	1.0	1.0	0.9996	0.2499	0.2499
2021-11-16	0.2499	1	1.0	1.0	1.0	0.2499	0.9996	0.2499
2021-11-17	0.2499	1	1.0	1.0	1.0	0.2499	0.2499	0.9996
2021-11-18	0.2499	1	1.0	1.0	1.0	0.2499	0.2499	0.2499
2021-11-19	-0.4965	-1	1.0	1.0	1.0	0.2499	0.2499	0.2499

weakening, or persisting across a short window while preserving realistic temporal ordering and avoiding look-ahead bias.

Once lagged features are introduced, any later forecasting improvement can be read as an effect of information that would actually have been available at the forecast date. This strengthens the methodological credibility of the thesis because it ties the behavioral signal to a realistic predictive setting rather than a retrospective one.

Through daily aggregation, distributional inspection, discretization, alignment, and lag construction, the raw sentiment data is transformed into a stable and interpretable behavioral signal suitable for transparent forecasting analysis.

3.3.2 MOMENTUM COMPONENT

Momentum captures whether Bitcoin is trading above or below its recent trend. Let Close_t denote the Bitcoin closing price at time t , and let $\text{MA}_{30,t}$ denote the 30-day moving average of the closing price. The raw momentum signal is defined as

$$\text{Momentum}_t = \frac{\text{Close}_t}{\text{MA}_{30,t}}.$$

Values above one indicate that price is trading above its recent average and therefore

exhibits positive directional pressure. Values below one suggest weaker momentum or downward pressure. Because this raw ratio is not naturally bounded, it is normalized to a 0–100 scale using an expanding min–max procedure. Momentum is included because fear and greed are also expressed through persistent directional movement. Sustained upward pressure often reflects confidence and speculative continuation, while weaker momentum or reversal pressure can reflect hesitation, stress, or deteriorating optimism.

Momentum also matters because trend itself can shape emotion. In crypto markets, upward continuation often strengthens optimism and encourages speculative participation, while weakening momentum can undermine confidence even before a full reversal occurs. Price trend therefore contributes to market state in its own right and deserves explicit representation inside the index.

3.3.3 VOLATILITY COMPONENT

Volatility is included because periods of fear are often associated with elevated uncertainty, sharp drawdowns, and unstable price behavior. Let the daily log return be defined as

$$r_t = \log \left(\frac{\text{Close}_t}{\text{Close}_{t-1}} \right).$$

The raw volatility signal is then computed as the rolling 30-day standard deviation of these log returns:

$$\text{Volatility}_t = \text{Std}(r_t)_{30}.$$

As with momentum, the raw volatility series is normalized to a 0–100 scale using an expanding min–max transformation. Because higher volatility is behaviorally associated with fear rather than greed, the normalized volatility contribution is inverted so that higher final values still correspond to more greed-like market conditions:

$$\text{VolatilityScore}_t = 100 - \text{ScaledVolatility}_t.$$

This inversion is conceptually important. In this thesis, volatility functions as a behavioral proxy for instability and market stress. When volatility rises sharply, the market is often entering a more fragile emotional regime. The inversion helps align the direction of the component with the overall interpretation of the index.

Volatility also helps the index distinguish between bullish continuation and unstable excitement. A market may be rising strongly while also becoming increasingly fragile. Including volatility helps the MFGI avoid reducing market state to direction alone. It adds information about whether that directional movement is occurring in a relatively orderly or relatively unstable environment.

3.3.4 SENTIMENT COMPONENT

The sentiment component of the MFGI uses the continuous daily mean sentiment signal introduced in Section 3.3.1. Unlike the discretized fear–neutral–greed classes used for lagged forecasting features, the index retains the continuous version of daily sentiment so that magnitude information is preserved. These daily values are also normalized through the same expanding min–max transformation, yielding a sentiment score between 0 and 100, where lower values represent stronger fear and higher values represent stronger greed.

This design creates an important distinction within the thesis. The discretized sentiment signal is used when a clear regime label is desirable for modeling and interpretation, while the continuous sentiment score is retained inside the MFGI to preserve a more graded measure of emotional intensity. In this sense, the sentiment component functions as the narrative layer of the index, capturing whether the broader information environment surrounding Bitcoin appears more optimistic, more fearful, or closer to neutral.

The sentiment component is especially important because it carries information that

price-derived measures alone cannot provide. Momentum and volatility describe what the market is doing. Sentiment helps describe how the broader information environment is framing that activity. This makes the index more behaviorally complete and better aligned with the way cryptocurrency markets actually evolve through narrative and reaction.

3.3.5 COMPOSITE AGGREGATION

After normalization, the three components are combined using a weighted linear aggregation:

$$\text{MFGI}_t = w_m \cdot \text{MomentumScore}_t + w_v \cdot \text{VolatilityScore}_t + w_s \cdot \text{SentimentScore}_t \quad (3.1)$$

Here, $w_m = 0.40$, $w_v = 0.30$, and $w_s = 0.30$. This weighting scheme gives slightly greater influence to momentum while still preserving meaningful contributions from volatility and sentiment. The weights are not tuned to maximize predictive accuracy. Instead, they are selected to preserve interpretability and to reflect a transparent behavioral design choice. Momentum captures trend persistence, volatility captures stress, and sentiment captures narrative tone. Their combination produces a single regime-level index rather than a highly optimized prediction feature.

This weighting choice also reflects a practical judgment about market structure. Momentum receives the largest weight because directional continuation is often one of the clearest visible expressions of speculative market behavior. In cryptocurrency settings, price trend can strongly shape expectations, attract attention, and reinforce crowd psychology. Volatility and sentiment remain highly important, but their roles are slightly more contextual. Volatility captures instability and regime fragility, while sentiment captures the narrative environment surrounding the market. This split therefore gives momentum a modest lead without overwhelming the other behavioral dimensions.

An equally weighted index would also have been reasonable, but the present weighting structure better reflects the role each component plays in the thesis. Momentum functions as

the most direct market-state signal, volatility contributes a measure of stress and instability, and sentiment contributes the informational and emotional layer. Because the goal is transparency and interpretability rather than optimized backtesting performance, the weighting scheme is presented as a design choice grounded in behavioral reasoning rather than as a tuned result.

These parameter choices should therefore be interpreted as transparent defaults rather than as claims of universal optimality, as in, a 30-day horizon is long enough to capture short-run market regime behavior while remaining responsive to changing conditions, and the selected weights preserve a balance between directional force, instability, and narrative tone. Different windows or weights would produce somewhat different index paths, but the purpose of the MFGI is not to maximize backtested fit through aggressive tuning. Its purpose is to provide a standard and interpretable summary of market state whose construction can be examined directly.

Table 3.2 shows sample rows from the resulting feature frame, including the normalized component scores, the final composite MFGI, and the lagged MFGI variables later used in the hybrid forecasting experiment.

Because the three raw components are measured on different numerical scales, each one must be transformed before aggregation. In this thesis, normalization is performed using an expanding min-max procedure rather than full-sample scaling. At time t , each component is scaled using only the minimum and maximum values observed up to time t , not values drawn from the full dataset. Formally, if x_t denotes a raw component value, then its scaled version is computed relative to $\min(x_1, \dots, x_t)$ and $\max(x_1, \dots, x_t)$. This design choice is important because it preserves realistic temporal ordering. A full-sample normalization would allow future extremes to influence earlier index values, which would introduce look-ahead bias into the construction of the MFGI. By contrast, the expanding-window approach ensures that every normalized score reflects only information that would have been available at that point in time. This strengthens the methodological transparency of the index and makes the

Table 3.2: Example rows showing the normalized momentum, volatility, and sentiment scores used to construct the MFGI, together with lagged MFGI features for later hybrid forecasting experiments.

Date	Close	Momentum _Score	Volatility _Score	Sentiment _Score	MFGI	MFGI _lag1	MFGI _lag2	MFGI _lag3
2021-09-09	46391.42	4.13	20.12	50.00	22.69	20.53	15.00	50.00
2021-09-10	44883.91	0.00	12.96	50.00	18.89	22.69	20.53	15.00
2021-09-11	45201.46	3.73	17.02	50.00	21.60	18.89	22.69	20.53
2021-09-12	46063.27	15.41	51.04	50.00	36.47	21.60	18.89	22.69
2021-09-13	44963.07	2.24	48.63	50.00	30.49	36.47	21.60	18.89
2021-09-14	47092.49	29.39	33.38	50.00	36.77	30.49	36.47	21.60
2021-09-15	48176.35	42.30	33.76	50.00	42.05	36.77	30.49	36.47
2021-09-16	47783.36	35.98	39.52	50.00	41.25	42.05	36.77	30.49
2021-09-17	47267.52	28.40	38.39	50.00	37.88	41.25	42.05	36.77
2021-09-18	48278.36	40.56	47.02	50.00	45.33	37.88	41.25	42.05
2021-09-19	47260.22	28.52	65.20	50.00	45.97	45.33	37.88	41.25
2021-09-20	42843.80	0.00	3.85	50.00	16.15	45.97	45.33	37.88

later forecasting analysis more credible, since the behavioral context signal is constructed under the same chronological discipline as the predictive models themselves. Figure 3.7 compares Bitcoin closing prices with the MFGI over the sample period. The two series do not move identically, and at some points they appear to diverge. This is expected because the MFGI is not a price-level measure. Instead, it provides a smoother, regime-oriented representation of the broader emotional and structural conditions surrounding price dynamics through momentum, volatility, and sentiment. As a result, Bitcoin can remain relatively high in absolute price terms while the MFGI declines if the market becomes more fragile, more volatile, or more pessimistic.

Several cyclical swings between lower and higher MFGI values are visible, supporting the interpretation of the index as a broad market-state descriptor rather than a short-horizon prediction target. In that sense, the MFGI helps characterize the behavioral regime in which daily price moves occur rather than attempting to replicate the price path itself.

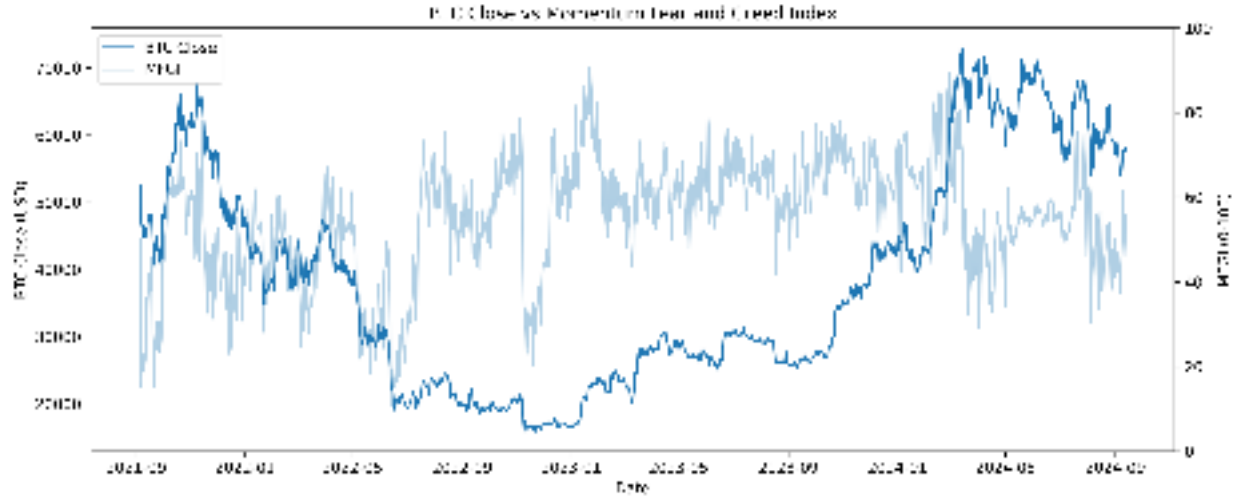


Figure 3.7: Bitcoin closing price and Momentum Fear and Greed Index (MFGI). The index summarizes broader market conditions rather than directly mirroring the raw price series.

3.3.6 COMPARISON WITH ALTERNATIVE.ME

To further evaluate the interpretive validity of the MFGI, the composite index was compared against the publicly available *alternative.me* Fear and Greed Index over their overlapping date range [2]. This comparison is not intended as an exact replication exercise. The proprietary benchmark appears to incorporate components such as social activity, market dominance, and search-interest information, but its full preprocessing rules and weighting logic are not made fully visible. Structural agreement between the two indices therefore provides useful external context.

Figure 3.8 overlays the MFGI with the *alternative.me* index over the matched comparison window.

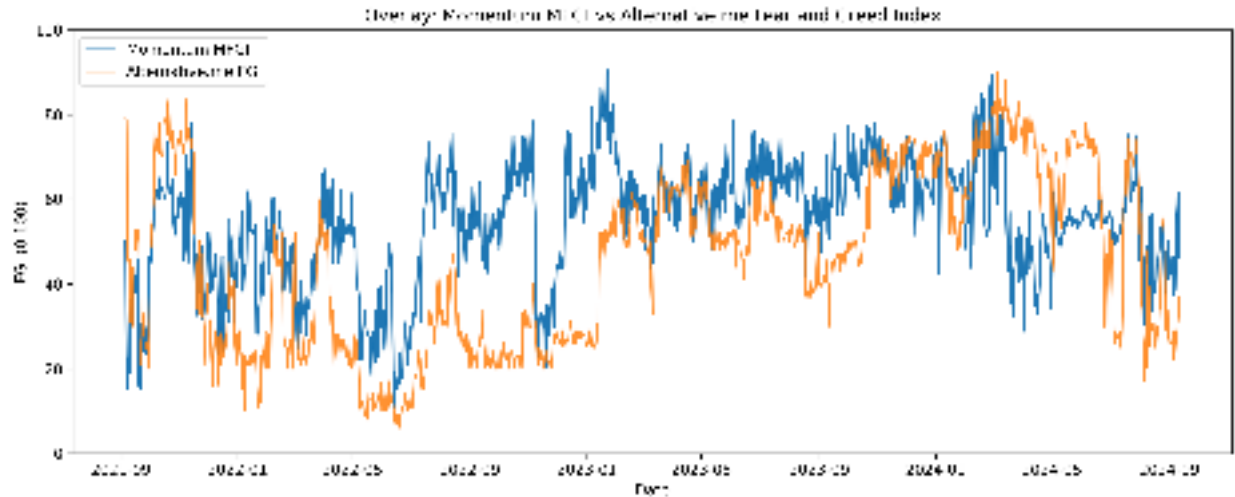


Figure 3.8: Overlay of the transparent Momentum Fear and Greed Index (MFGI) and the proprietary *alternative.me* Fear and Greed Index over their overlapping historical window.

The two indices do not coincide perfectly at the day-to-day level, but they display meaningful structural alignment across major swings. Broad periods of rising optimism, declining fear, and renewed stress appear in both series, even though their amplitudes and short-run responsiveness differ. This is consistent with the methodological distinction between them. The *alternative.me* index reflects a broader proprietary input set, whereas the MFGI relies only on explicitly defined momentum, volatility, and aggregated news sentiment. Divergence therefore reflects differences in construction.

The structural alignment becomes easier to interpret when major market events are considered directly. During 2022, cryptocurrency markets experienced severe trust shocks and broad declines in confidence, including the wider stablecoin crisis and the bankruptcy of FTX [13, 14]. In Figure 3.8, both the MFGI and the *alternative.me* benchmark spend substantial time in lower ranges during this broader period of fear and instability. Both indices appear to capture the same broad market regime even when their exact daily paths differ.

The 2022 period is especially valuable for interpretation because it tests whether the transparent index can register prolonged market distress rather than only isolated daily

shocks. A meaningful fear-and-greed measure should respond to sustained breakdowns in confidence, not just short bursts of volatility. In this case, the MFGI remains subdued for extended stretches of the crisis period, which supports the view that it is capturing more than temporary noise. The broad alignment with the proprietary benchmark strengthens that interpretation.

A second useful period for interpretation appears during 2023, when sentiment was shaped by both regulatory pressure and more favorable legal developments. Enforcement actions against major exchange actors such as Binance contributed to market uncertainty [17], while the Grayscale court decision altered expectations surrounding the regulatory future of Bitcoin-related financial products [16]. These mixed forces help explain why both indices show broad directional sensitivity to changes in market mood while still differing in short-run amplitude. A transparent index should not be expected to match a proprietary one point for point, especially when the proprietary benchmark draws on a wider and less visible input set.

This 2023 period also illustrates an important methodological point. Structural agreement matters more than exact daily coincidence. The purpose of the MFGI is to capture broad market-state movement through visible components. It is not designed to reconstruct every fluctuation in a proprietary benchmark whose internal logic cannot be fully inspected. If both indices rise during improving sentiment environments and weaken during stress episodes, that structural correspondence is analytically meaningful even when the day-to-day paths differ.

The comparison window in early 2024 is especially useful because several closely spaced events influenced market sentiment in rapid succession. These include the false social-media report suggesting Bitcoin ETF approval [3], the formal approval of spot Bitcoin ETFs in the United States [15], and the approach of the 2024 Bitcoin halving event [12]. Together, these developments contributed to alternating bursts of optimism, uncertainty, and renewed speculative attention. Both the MFGI and the *alternative.me* index respond to this changing environment, though not with identical timing or intensity. The two indices appear to track similar market regimes while remaining methodologically distinct.

Early 2024 is also helpful because it contains both confusion and optimism in close proximity. The false ETF report generated a temporary shock of uncertainty and informational disorder. The actual ETF approval generated renewed enthusiasm and stronger expectations about institutional acceptance. The halving added another layer of speculative anticipation. These events create a demanding comparison environment because market emotion shifts quickly and for different reasons within a relatively short span of time. The broad responsiveness of both indices during this period suggests that the MFGI remains sensitive to meaningful changes in regime even without access to a proprietary feature set.

This comparison is important because it clarifies the tradeoff at the center of the thesis. The AFGI offers visibility and market familiarity, but limited interpretive transparency. The MFGI gives up some of the proprietary breadth of the commercial benchmark in exchange for analytical clarity. It makes each component visible, each transformation traceable, and each weighting choice intelligible. The goal is to produce a transparent approximation of broad market-state behavior that remains useful for research and forecasting.

The comparison also strengthens the role of the MFGI within the broader thesis framework. A transparent behavioral index is more persuasive when it does two things at once: it remains interpretable on its own terms, and it still captures market-state movement that resembles a widely referenced benchmark. The overlay suggests that the MFGI satisfies both conditions. It remains fully inspectable, and it still reflects the broad emotional contours that make fear-and-greed style indicators relevant to practitioners and observers.

From an interpretive standpoint, the overlay matters for two reasons. First, it suggests that a transparent framework can still capture much of the same regime-level information that makes proprietary sentiment indices widely referenced in practice. Second, it supports the thesis emphasis on interpretability. Because every movement in the MFGI can be traced to observable components, the index functions as a market mood score and an auditable research instrument.

The sentiment preprocessing pipeline transformed raw headline data into structured

daily signals through aggregation, distributional analysis, discretization, alignment, and lag construction. The MFGI then extended that behavioral foundation by integrating sentiment with momentum and volatility into a transparent composite index. Together, these components provide the contextual layer missing from price-only forecasting and prepare the ground for the sentiment-aware model comparisons examined in the next chapter.

CHAPTER 4

IMPLEMENTATION AND ANALYSIS OF MODELS

This chapter evaluates whether adding behavioral context from the Momentum Fear and Greed Index (MFGI) improves LSTM-based Bitcoin forecasting relative to a comparable price-only model. Rather than introducing a new family of forecasting architectures, the analysis keeps the underlying sequence model broadly consistent so that differences in behavior and performance can be interpreted primarily in terms of informational input rather than architectural change. The comparison therefore focuses on two settings: a price-only LSTM that uses historical Bitcoin closing prices alone, and a hybrid LSTM that augments the same price history with lagged MFGI features.

Both models are trained on the same daily forecasting task, use the same chronological train–test structure, and preserve realistic temporal ordering throughout preprocessing and input construction. This controlled design makes it possible to evaluate whether the transparent behavioral signal developed in Chapter 3 contributes useful information beyond price history alone. In this sense, the chapter addresses the forecasting side of the second research question by moving from framework construction to direct empirical comparison.

To evaluate predictive performance consistently, both models are assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Squared Error (MSE). Recall that MAE measures the average absolute deviation between predicted and observed prices in the original Bitcoin price scale, making it easy to interpret in practical terms; RMSE penalizes larger errors more heavily and is therefore especially informative in volatile markets

where occasional large misses matter disproportionately; MSE, as the squared counterpart of RMSE, provides the raw squared-error quantity and makes relative differences in larger forecast mistakes more visible. Taken together, these three measures provide a balanced view of model performance, capturing both average error magnitude and sensitivity to extreme deviations.

Another reason for using all three metrics is that they help distinguish different kinds of forecasting weakness. Two models can have similar average error while differing meaningfully in how often they miss badly during volatile episodes. A model with slightly lower MAE but substantially lower RMSE is useful because it suggests better control over the larger misses that matter most during unstable market periods. In a setting such as Bitcoin forecasting, where regime shifts and abrupt accelerations are central to the problem, this distinction is especially important.

4.1 PRICE-ONLY LSTM

The price-only LSTM serves as the baseline nonlinear sequence model. Its role is to establish how much structure can be learned from historical Bitcoin prices alone before any behavioral features are introduced. The model therefore isolates the predictive content of price history without access to sentiment, momentum, volatility, or any external contextual variables. Although this model is also price-only, it is distinct from the benchmark LSTM reported in Chapter 2. The Chapter 2 LSTM was used in the broader price-only comparison that addressed RQ1, whereas the present baseline is constructed on the aligned modeling dataset used for the controlled RQ2 comparison, with training data from September 9, 2021 through February 5, 2024 and a held-out test period from February 6, 2024 through September 12, 2024.

That distinction matters because the Chapter 2 LSTM was designed to answer a broader benchmarking question: how does a price-only LSTM compare with naïve persistence and

ARIMA within a pure price-only framework? The Chapter 4 LSTM serves a different purpose. It is the baseline inside the final modeling environment used for the hybrid comparison. The two models are related, yet they serve different experimental purposes. They belong to different stages of the thesis and answer different research questions.

Let p_t denote the Bitcoin closing price on day t . The model is trained on rolling input windows of length $N = 60$, so that each example uses the previous sixty daily closing prices to predict the next day’s closing price. Formally, each input–target pair is defined as

$$X_t = [p_{t-60}, p_{t-59}, \dots, p_{t-1}], \quad y_t = p_t.$$

As in Chapter 2, the 60-day lookback window is used as a practical compromise between short-run responsiveness and medium-term sequence memory. A shorter lookback would make the network more reactive to local fluctuations, but would also reduce its access to broader trend structure. A longer lookback would provide more context, but at the cost of greater dimensionality and reduced sensitivity to recent movement. In this implementation, the 60-day choice therefore remains a reasonable and interpretable benchmark rather than an attempt to optimize the architecture exhaustively.

The lookback window is also important because it determines what kind of temporal structure the model is actually able to learn. With a 60-day window, the network can observe multiweek directional movement, volatility buildup, and short-run reversals that unfold over more than a few trading days. This is useful in Bitcoin markets, where price behavior often evolves through bursts of momentum rather than through isolated daily shocks. A very short lookback might produce a more reactive network, but it would also push the model toward very local behavior and increase the risk that it learns noise rather than broader structure.

Before sequence construction, the closing price series is normalized using a MinMaxScaler fitted only on the training portion of the data. This maps the input values to the interval $[0, 1]$ and stabilizes neural network training while preserving causal validity by preventing information from the test period from influencing preprocessing.

This preprocessing decision is part of the broader temporal discipline of the chapter. All transformations are designed to respect the direction of time. The model is allowed to learn from past observations and past scaling ranges, but not from future information embedded in the held-out period. This is especially important in financial forecasting, where preprocessing mistakes can quietly leak future information into the model and make forecast performance look stronger than it really is.

The baseline architecture is intentionally simple, to provide a clean nonlinear benchmark against which the sentiment-aware extension can be compared. A simpler architecture helps keep the focus on information content rather than network depth, hyperparameter search, or architectural specialization. The network consists of one LSTM layer with 64 hidden units, one dropout layer with rate 0.2, and one dense output layer that produces the next predicted closing price. Training uses the Adam optimizer with mean squared error loss. Adam is especially suitable here because it adapts the learning rate during training and tends to perform well on noisy optimization problems, which is useful in financial time series where local gradient behavior can be unstable. Early stopping is applied to reduce overfitting by restoring the weights associated with the lowest validation loss, and shuffling is not used during training so that temporal structure is preserved.

Figure 4.1 shows the price-only LSTM predictions against the observed Bitcoin closing prices over the held-out test period.

The figure shows that the model captures broad directional movement, but its predictions remain visibly smoother than the true series. This behavior is typical of squared-error sequence models applied to volatile financial data: the network learns general temporal structure, yet tends to underrepresent abrupt reversals and sharper local fluctuations. In practical terms, the model behaves as a reasonable nonlinear baseline, but it still reflects the broader limitation identified earlier in the thesis: price-only forecasting struggles most when market behavior becomes unstable, emotionally amplified, or regime-dependent.

This pattern is also reflected in the model’s evaluation metrics. Under the final test setup,

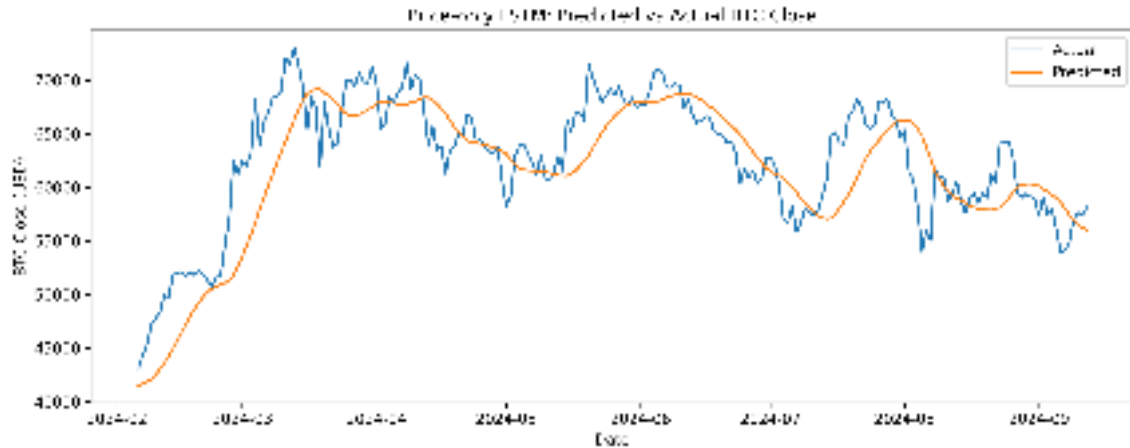


Figure 4.1: Price-only LSTM predictions versus actual Bitcoin closing prices over the held-out test period.

the price-only LSTM achieved a Mean Absolute Error (MAE) of 2943.08 USD, a Root Mean Squared Error (RMSE) of 3812.38 USD, and a Mean Squared Error (MSE) of 14,534,245.23 USD². These values indicate that the model captures some meaningful nonlinear temporal structure, but still produces errors that remain substantial in absolute terms. The RMSE being notably larger than the MAE further suggests that the model occasionally misses more sharply during larger market moves, which is consistent with the visual smoothing seen in Figure 4.1.

Interpreting these metrics together helps clarify the role of the baseline model. The somewhat higher MAE, RMSE, and MSE in this chapter should be understood in light of the different aligned dataset and held-out time period used for the controlled RQ2 comparison. Even so, the results still reinforce the point that nonlinear sequence modeling cannot fully overcome the volatility and abruptness of Bitcoin price behavior when the model is restricted to price history alone. These results also confirm something already suggested by the earlier chapters: model flexibility helps, yet flexibility alone is not enough. A price-only sequence model can learn broad movement and local patterning, yet it still operates inside a narrow information space. When Bitcoin reacts to market stress, changes in mood, or broader narrative developments, the network can only infer those effects indirectly through price

history. That leaves it vulnerable during the exact periods that matter most for regime-aware forecasting.

4.2 HYBRID LSTM WITH LAGGED MFGI FEATURES

The hybrid LSTM extends the baseline model by incorporating lagged values of the Momentum Fear and Greed Index alongside historical Bitcoin prices. The purpose of this extension is to determine whether the transparent behavioral signal developed in Chapter 3 contributes information that is not already contained in price history alone. Rather than replacing the price-only architecture with a substantially different framework, the hybrid model preserves the same general network design and training logic so that the comparison remains interpretable.

The target variable remains the daily Bitcoin closing price, but the feature space is expanded to include lagged MFGI values. Specifically, the hybrid model uses the daily closing price together with MFGI_{t-1} , MFGI_{t-2} , and MFGI_{t-3} . These lagged values are used instead of contemporaneous values so that each forecast depends only on information that would have been available before the prediction date. This is methodologically important because the purpose of the hybrid model is to test whether behavioral context can contribute useful information under realistic temporal ordering, not whether same-day sentiment can artificially inflate forecasting performance.

The use of three lagged MFGI values rather than only one is also meaningful. A single lag would provide only the most immediate behavioral snapshot. Multiple lags allow the model to see short-run persistence in market state and to infer whether fear or greed is strengthening, fading, or remaining stable across several days. In that sense, the hybrid model receives a brief recent history of behavioral context rather than one isolated regime signal. In simplified form, each time step s in the hybrid input window is represented as

$$x_s = (p_s, \text{MFGI}_{s-1}, \text{MFGI}_{s-2}, \text{MFGI}_{s-3}),$$

where $s = t - 60, \dots, t - 1$. The full multivariate input sequence is therefore

$$X_t = [x_{t-60}, x_{t-59}, \dots, x_{t-1}],$$

with target

$$y_t = p_t.$$

In practice, the implementation uses the aligned daily modeling frame produced after MFGI construction. This means that each timestep contains both market price information and lagged behavioral regime information derived from the transparent index framework. The hybrid model is therefore still forecasting price, but it does so with access to a broader information set that includes regime-aware behavioral context.

This aligned modeling frame is one of the most important features of the chapter. By the time the data reaches the hybrid LSTM, the MFGI has already been constructed, normalized, aligned to price, and lagged appropriately. That means the hybrid model operates on a clean daily structure in which price and behavioral context are temporally compatible. This is what makes the comparison persuasive. The hybrid model receives a clearly defined behavioral signal that has been prepared for use in a realistic forecasting setting.

The hybrid architecture remains intentionally close to the baseline case. It also uses one LSTM layer with 64 hidden units, one dropout layer with rate 0.2, and one dense output layer. The primary difference is therefore not architectural depth, but informational scope. Because the hybrid model receives multiple features per timestep rather than only price, its input tensor is multivariate, but the overall modeling logic remains comparable to that of the baseline. As in the price-only case, the data is split chronologically, scaling is fit only on the training portion, the Adam optimizer is retained, and early stopping is used during training. This keeps the comparison controlled and makes it easier to attribute any performance differences to the added behavioral signal rather than to unrelated architectural changes.

Figure 4.2 shows the hybrid LSTM predictions against the observed Bitcoin closing prices over the held-out test period. At the implementation level, the hybrid model behaves similarly to the baseline in overall form, but it operates with a broader input space. The model still produces a smooth sequence approximation, yet it does so using both past prices and lagged behavioral context. Visually, the hybrid forecasts tend to remain slightly closer to the observed series across several intervals, suggesting that the added regime-aware information helps the network adjust more effectively to changing market conditions.

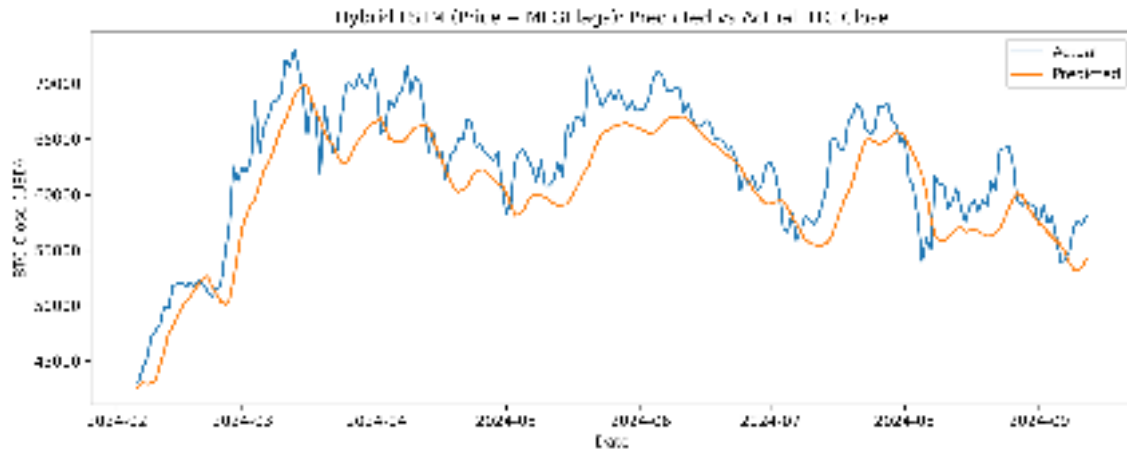


Figure 4.2: Hybrid LSTM predictions versus actual Bitcoin closing prices using price and lagged MFGI inputs.

This improvement is also visible in the model’s evaluation metrics. Under the same test setup, the hybrid LSTM achieved an MAE of 2930.37 USD, an RMSE of 3589.96 USD, and an MSE of 12,887,839.72 USD². Thus, the hybrid model outperformed the price-only baseline on all three error measures. The gain is not dramatic, but it is consistent. This matters because it indicates that the MFGI contributes nonredundant information rather than merely restating what is already contained in the price series.

The relative drop in RMSE is especially meaningful. Because RMSE weights larger forecast misses more heavily, its improvement suggests that the hybrid model is somewhat better at managing the periods in which price-only forecasting tends to struggle most. The reduction in MSE reinforces this interpretation by showing that the squared cost of larger errors is lower

once lagged behavioral information is added. The MAE improvement is smaller, but still directionally consistent, which suggests that the hybrid model provides modest improvement not only on extreme misses, but also across average forecast performance.

The modest scale of the improvement is also important. The hybrid model does not transform Bitcoin into an easily predictable asset, nor does it remove the instability inherent in short-horizon cryptocurrency forecasting. Instead, it provides incremental predictive value, which is exactly the kind of result one would expect from a contextual behavioral signal rather than a direct forecasting oracle. In this sense, the results support the claim that sentiment-aware features enrich the informational environment of the model without eliminating the deeper uncertainty of the market.

The hybrid model is useful because it sharpens the model’s access to regime information. Price history tells the network what the market has done. Lagged MFGI features give it an additional signal about the broader emotional and structural environment in which that behavior is occurring. This does not allow the network to escape uncertainty, but it does help it situate price movement inside a wider context than raw closing prices alone can provide.

4.3 COMPARATIVE ANALYSIS (RQ2)

Taken together, the two implementations define the core forecasting comparison underlying the second research question. The baseline model asks how much can be learned from historical Bitcoin prices alone, while the hybrid model asks whether the same forecasting architecture benefits from the addition of lagged behavioral context captured by the MFGI.

This comparison was designed to remain as controlled as possible. Both models operate on the same target variable, use the same rolling window length, preserve chronological train–test ordering, rely on the same broad LSTM structure, and are trained under the same general optimization logic. The main substantive difference is the inclusion of the lagged MFGI features in the hybrid setting. As a result, differences in forecasting behavior can be

interpreted primarily in terms of the informational role of the added behavioral signal rather than in terms of unrelated architectural changes.

For clarity, Table 4.1 summarizes the comparative performance of the two LSTM configurations. The empirical pattern is clear. The transparent MFGI does improve LSTM forecasting relative to the price-only baseline, but the improvement is modest rather than transformative. This suggests that the behavioral signal contributes useful contextual information beyond price history alone, while also confirming that cryptocurrency forecasting remains structurally difficult even when richer inputs are added. The MFGI is best understood as a regime-aware contextual feature that slightly sharpens the model’s understanding of market conditions.

Table 4.1: Comparison of price-only and hybrid LSTM forecasting performance. Lower values indicate better predictive performance.

Model	MAE (USD)	RMSE (USD)	MSE (USD ²)
Price-only LSTM	2,943.08	3,812.38	14,534,245.23
Hybrid LSTM + lagged MFGI	2,930.37	3,589.96	12,887,839.72

Relative to the price-only baseline, the hybrid model reduces MAE by about 0.43%, RMSE by about 5.83%, and MSE by about 11.33%. Because the improvement is larger in RMSE and MSE than in MAE, the added MFGI signal appears to help more with larger forecast misses than with average day-to-day error. The metric pattern also helps clarify the nature of the gain. If improvement had appeared in only one metric, the result would be more ambiguous. Instead, the hybrid model improves MAE, RMSE, and MSE simultaneously. This does not imply a breakthrough in predictive power, but it does strengthen the case that the additional behavioral input is genuinely helpful. The largest relative benefit appears in the squared-error measures, which suggests that the hybrid model is especially useful in reducing the severity of larger forecast misses, even if it does not eliminate them.

This point is important conceptually. In volatile financial markets, modest but consistent improvement can be more believable than large headline gains. A dramatic reduction in

forecast error might raise concerns about overfitting, hidden leakage, or dependence on a particularly favorable window. By contrast, a smaller but stable improvement across all three metrics is more consistent with the claim that the model is benefiting from a useful but limited contextual signal.

It is also worth emphasizing that this result fits the broader logic of the thesis. Chapter 2 showed that price-only forecasting eventually reaches an informational boundary. Chapter 3 developed a transparent index intended to represent broader market state. The present chapter brings those ideas together empirically. Once the model is given access to a structured behavioral signal, performance improves. The improvement is limited, but it is directionally coherent and methodologically plausible.

A visual comparison between the two forecasting approaches further reinforces this point. Figure 4.3 shows the price-only and hybrid forecasts against the observed Bitcoin series. This visual pattern is consistent with the numerical results. Both models remain constrained by the volatility and abruptness of the underlying market, but the hybrid model shows a slightly stronger ability to remain aligned with the observed series. That visual closeness is reflected quantitatively in the lower error metrics reported above.

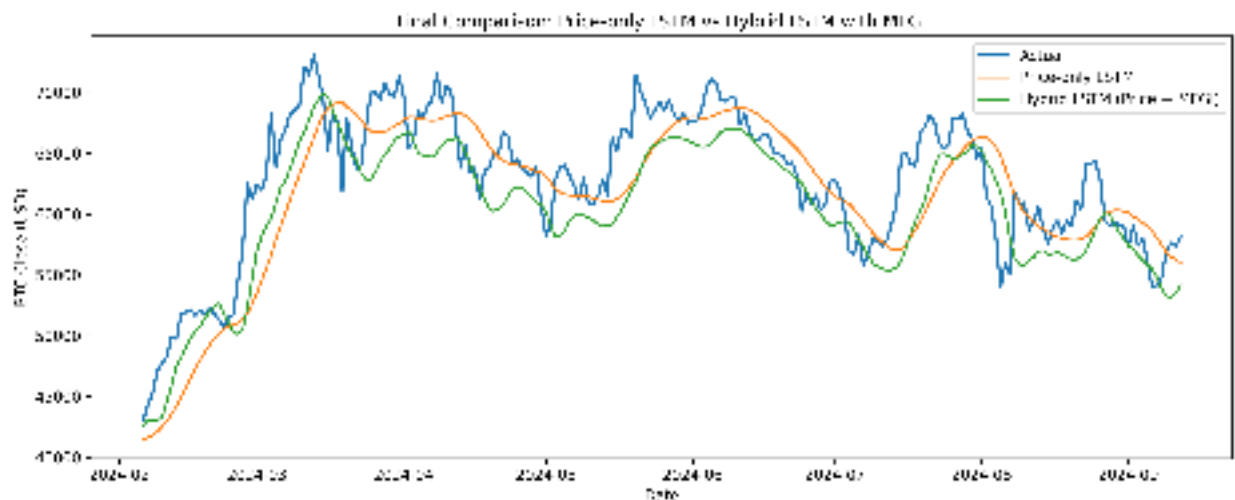


Figure 4.3: Comparison of price-only and hybrid LSTM forecasts against actual Bitcoin closing prices. The hybrid model remains smoother than the observed series, but generally tracks the target slightly more closely than the price-only baseline.

RQ2 Answer: Incorporating lagged MFGI features into the LSTM improves forecasting performance beyond what is achieved using historical price data alone, but the improvement is incremental rather than dramatic. This indicates that transparent sentiment-aware context adds useful information to the model, while also reinforcing that short-horizon cryptocurrency forecasting remains limited by broader market uncertainty.

This result also connects back to the broader comparison between the MFGI and the Alternative.me index developed in Chapter 3. MFGI provides a transparent and interpretable alternative to a widely used but opaque proprietary benchmark. The present chapter extends that contribution by showing that the transparent index is also operationally useful within a controlled forecasting framework. In that sense, the thesis constructs a transparent sentiment measure and demonstrates that it can function as a meaningful forecasting input within a carefully designed sequence model.

The broader implication is that forecasting improvements in cryptocurrency markets may depend less on endlessly increasing architectural sophistication and more on enriching the information available to the model in a careful and interpretable way.

This chapter therefore completes the forecasting side of the second research question. The next and final chapter turns to the broader conclusion of the thesis by synthesizing what the price-only benchmarks, the transparent MFGI framework, and the hybrid forecasting comparison together imply about transparency, interpretability, and the future of sentiment-aware cryptocurrency forecasting.

CHAPTER 5

CONCLUSION

This thesis examined whether a transparent and interpretable sentiment-aware framework could improve cryptocurrency forecasting beyond price-only methods while preserving methodological clarity. The analysis moved from price-only benchmark modeling, to the construction of a transparent behavioral signal, to a controlled comparison between a price-only LSTM and a hybrid LSTM incorporating lagged Momentum Fear and Greed Index (MFGI) features. Across these stages, one result remained clear: cryptocurrency forecasting is constrained not only by model choice, but also by the limits of the information given to the model.

The central contribution of *Momentum AI* is the demonstration that a transparent behavioral framework can expand that information set in a meaningful and interpretable way. Historical prices alone contain useful short-run structure, but they do not fully capture the emotional, narrative, and regime-driven forces that shape digital asset markets. The MFGI was developed in response to that limitation as a transparent composite signal built from momentum, volatility, and aggregated news sentiment. In that sense, the project was never only about producing a slightly different forecast. It was also about asking what kind of market information deserves to be represented explicitly in a forecasting system.

5.1 OVERALL SYNTHESIS OF FINDINGS

The findings of the thesis support three broad conclusions. First, price-only forecasting remains informative but limited. The results in Chapter 2 showed that Bitcoin price history contains short-run persistence and some usable temporal structure, but that this structure is highly regime dependent. The naïve benchmark performed strongly in the shared evaluation window, indicating that local momentum can dominate short-horizon forecasting. At the same time, the weaker performance of ARIMA and the smoothing behavior of the price-only LSTM showed that increasing model sophistication alone does not overcome the informational limits of historical price data. More complexity can change how a model responds to price history, but it does not automatically solve the deeper problem that price alone may not carry enough context during unstable periods.

Second, behavioral context adds value when it is structured carefully. Chapter 3 developed the argument that cryptocurrency markets are shaped by price history, collective attention, emotional reaction, and regime-sensitive market behavior. This motivated the construction of the MFGI as a transparent alternative to opaque proprietary sentiment indicators. Chapter 4 then showed that incorporating lagged MFGI values into the hybrid LSTM improved performance modestly but consistently relative to the price-only LSTM. The gain was limited, but it appeared across the reported error metrics, suggesting that the added behavioral information was genuinely useful. The result is important not because it transforms forecasting performance, but because it shows that even a relatively simple behavioral signal can contribute information that price-only inputs leave out.

Third, transparency remained compatible with usefulness. One of the broader methodological concerns of the thesis was whether an open and interpretable index could still capture behaviorally meaningful market structure without relying on proprietary or black-box design choices. The structural comparison between the MFGI and the *alternative.me* Fear and Greed Index showed that the transparent framework was capable of reflecting many of the same broad swings in market mood while preserving full visibility into its components,

transformations, and weighting logic. This indicates that interpretability and relevance can coexist. For research purposes, that matters because a transparent index can be examined, questioned, and improved in ways that a closed benchmark cannot.

Taken together, these findings support a broader conclusion about forecasting design. Within the scope of this thesis, the strongest gains came from clarifying what price-only models can and cannot explain, then expanding the information available to the model through a structured behavioral signal. In that sense, the project contributes to forecasting methodology as well as forecasting performance. It suggests that better forecasting does not always begin with a more complex architecture. Sometimes it begins with a better representation of market context.

5.2 ANSWER TO RQ3

The final research question concerns whether a transparent sentiment-aware framework can retain interpretability without sacrificing practical usefulness. The results of this thesis support an affirmative but carefully qualified answer.

RQ3 Answer: A transparent sentiment-aware framework can preserve interpretability while still providing meaningful analytical and forecasting value. Although the transparent MFGI does not produce dramatic gains or exactly replicate a proprietary benchmark, it captures broad market-state information, contributes modestly to forecasting performance, and remains transparent in a way that opaque systems do not.

This conclusion matters because it challenges a common assumption in financial modeling and data-driven forecasting: that usefulness necessarily requires opacity. In many practical settings, especially in markets shaped by speculation and rapid information flow, black-box indicators are often treated as inherently more powerful than open ones. The findings of this thesis do not support that assumption in a simple way. The MFGI is fully transparent and

interpretable, yet it still functions as a meaningful contextual signal within the forecasting framework and shows structural alignment with a widely referenced proprietary alternative.

At the same time, this thesis does not establish that a transparent MFGI-based forecasting system is superior to a forecasting system built directly on a proprietary AFGI. A direct head-to-head forecasting comparison of that kind was not conducted. The present results instead show that a transparent index can remain practically useful without depending on hidden formulas or undisclosed transformations. That is an important distinction. The claim here is not that transparency wins in every case, but that transparency does not prevent a model from being informative.

This also does not mean that transparency guarantees better forecasting. The hybrid LSTM remained imperfect, and the MFGI did not transform Bitcoin into an easily predictable asset. Transparency also introduces a possible gaming risk. Because the construction of the signal is visible, market participants could in principle attempt to influence some observable inputs or strategically react to the published logic of the index. In the present framework, that risk is reduced by the use of multiple components, aggregation, and lagged inputs rather than a single easily targeted signal, but it cannot be eliminated entirely.

More specifically, transparency can create an opening for strategic behavior if market participants know which kinds of observable signals are being incorporated into the index. A malicious actor would not need to control the entire market to exploit that visibility. It could be enough to influence some public-facing inputs, amplify particular narratives, or encourage reactions around signals that traders already associate with bullish or bearish conditions. In that case, the transparent structure of the index could make some parts of the forecasting framework easier to anticipate or strategically pressure, even if it does not make the forecast itself directly controllable. This does not undermine the value of transparency, but it does show that transparency introduces a tradeoff between interpretability and exposure to possible behavioral manipulation.

The more important result is that transparency and usefulness coexisted. The signal

could be understood clearly, questioned openly, and still contribute something empirically valuable. A transparent model or index can be audited, refined, and interpreted, which makes it especially valuable for research and, in many settings, more responsible for practical use. In a market where opaque indicators are often accepted without much scrutiny, that openness is itself a meaningful contribution.

5.3 LIMITATIONS

The MFGI is intentionally limited in scope. Because it is built only from explicitly documented components, momentum, volatility, and aggregated news sentiment, it does not include every source of market mood, such as derivatives positioning, order-book behavior, on-chain flows, broader social-media propagation, or search-interest dynamics. This strengthens transparency, but it also restricts the information available to the index. The framework is therefore useful as a focused behavioral signal, not as a complete summary of all forces acting on the market.

The use of lagged features introduces a further tradeoff. Lagging is necessary to preserve realistic temporal ordering, but it also reduces immediacy. If sentiment shifts extremely quickly around a market shock, some of that change may already be partially reflected in price before the lagged feature contributes strongly to the forecast. This means that the framework is better suited to capturing evolving market state than to reacting instantly to the first moments of a sudden shock.

Another limitation concerns the structure of the underlying sentiment data itself. The sentiment component of the MFGI is based on aggregated Bitcoin-related news headlines, which means that the index reflects one important channel of public information rather than the full emotional landscape of the cryptocurrency market. News sentiment can capture narrative direction and major informational shifts, but it may miss the intensity and speed of reactions that emerge first through retail discussion forums, short-form social media, or

exchange-specific communities. This matters because cryptocurrency markets often process sentiment through multiple channels at once, and these channels do not always move together.

The thesis is also limited by its emphasis on a specific asset, a specific date range, and a specific forecasting horizon. Bitcoin is the most visible and institutionally recognized cryptocurrency, which makes it a sensible choice for analysis, but it is not automatically representative of the broader digital asset ecosystem. Smaller assets often exhibit stronger retail sensitivity, thinner liquidity, sharper swings, and different sentiment dynamics. In addition, the present analysis focuses on daily short-horizon forecasting over the study period examined here. Different conclusions might emerge under intraday forecasting, longer-term horizon settings, or alternative historical windows where broader trend structure dominates more of the signal.

A related limitation is that the thesis is restricted to Bitcoin alone. Bitcoin is the most visible and data-rich cryptocurrency, which makes it an appropriate starting point for a study of this kind, but it is only one part of the broader digital asset ecosystem. The behavior of Ethereum, smaller altcoins, or more thinly traded tokens may differ substantially in terms of liquidity, retail sensitivity, sentiment transmission, and volatility structure. As a result, the conclusions of this thesis should be interpreted primarily as Bitcoin-specific findings rather than as claims that automatically extend to all cryptocurrency markets. This is especially important because different assets may respond to public narrative, speculation, and volatility in uneven ways.

The evaluation window itself also matters. Forecasting results in cryptocurrency markets are highly regime dependent, and the relative strength of any model depends partly on the structure of the period being studied. A window dominated by strong trend continuation may favor persistence-based approaches, while a more volatile and sentiment-reactive period may increase the usefulness of broader behavioral context. For that reason, the reported results should be understood as findings grounded in the specific market conditions examined here, not as universal rankings that would hold identically across every period.

5.4 FUTURE WORK

Several directions remain open for future research. One natural extension would be to incorporate additional behavioral sources beyond news sentiment, such as social-media discussion, search-interest dynamics, on-chain indicators, or derivatives-based measures of market stress. These additions could enrich the information set further, though they would need to be integrated carefully to preserve transparency. The challenge will be to expand coverage without turning the framework into a collection of loosely connected signals that are difficult to interpret.

Many of these future directions follow directly from the limitations of the present study. Because the analysis is confined to Bitcoin, a specific date range, and a daily short-horizon forecasting setting, an important next step is to test the framework across additional assets, broader historical windows, and alternative forecasting horizons. Doing so would help determine which findings are specific to Bitcoin and which reflect more general properties of transparent sentiment-aware forecasting. Future work should also examine whether the risks introduced by transparency, including the possibility of strategic pressure on visible inputs, change when the signal is built from a broader and more diversified set of market-state indicators.

Another direction would be to test the framework across different assets and time horizons. The present thesis focuses on Bitcoin and short-horizon daily forecasting, but other cryptocurrencies may respond differently to sentiment and behavioral context. Regime-specific evaluation across bull markets, bear markets, and volatility shocks could also clarify when transparent behavioral signals matter most. This would help separate asset-specific behavior from broader forecasting patterns.

Future work could also include a direct forecasting comparison between transparent MFGI-based models and models built on a proprietary benchmark such as the *alternative.me* Fear and Greed Index, where data availability permits. That would clarify whether the value of transparency in this setting is primarily interpretive, forecasting-related, or both. It would

also provide a clearer answer to how much performance, if any, is traded away when opacity is removed from the design.

Future work could also explore alternative transparent architectures. The present study intentionally used a restrained LSTM design in order to keep the comparison interpretable. Subsequent research could test whether more advanced but still understandable sequence models can extract stronger value from behavioral features without sacrificing clarity. This would be especially useful for determining whether the main bottleneck lies in the signal itself, the model architecture, or the interaction between the two.

Another promising direction would be to explore more adaptive versions of the MFGI itself. The present index uses fixed component weights in order to preserve interpretability, but future research could investigate whether transparent regime-sensitive weighting schemes improve the index’s usefulness while keeping its construction understandable. This could help clarify whether the relative importance of momentum, volatility, and sentiment changes across different market conditions.

A further extension would be to examine event-specific responsiveness more directly. The present thesis interprets the MFGI around major market episodes such as the TerraUSD collapse, the FTX bankruptcy, regulatory actions, ETF-related developments, and the Bitcoin halving period. A future study could evaluate this relationship more systematically by pairing event windows with formal changes in the index and comparing those shifts across different market conditions. This would strengthen the connection between transparent sentiment measurement and concrete market interpretation.

APPENDIX A

USE OF ARTIFICIAL INTELLIGENCE

Generative artificial intelligence tools were used in a limited and supporting capacity during the development and writing of this Independent Study thesis. Their use was restricted to brainstorming, clarification of technical concepts, coding and debugging assistance, organizational support, and revision of drafted writing for clarity, grammar, and readability. At no point were AI tools used to fabricate data, generate experimental results, or determine the final conclusions of the thesis.

During the research and implementation process, ChatGPT (OpenAI GPT-5) was used as an interactive assistant when clarifying forecasting concepts, interpreting technical documentation, debugging code behavior, and improving the structure of written explanations. These interactions functioned in a manner similar to consulting technical documentation, programming forums, or a writing support tool. Most prompts consisted of targeted questions about time-series forecasting, sentiment preprocessing, model behavior, evaluation metrics, and LaTeX writing or formatting issues encountered during the development of the project.

For example, AI tools were consulted while working through aspects of the forecasting pipeline used in *Momentum AI*, particularly when clarifying concepts related to ARIMA modeling, LSTM sequence construction, rolling-window input design, and evaluation metrics such as MAE, RMSE, and MSE. These interactions were used to better understand the behavior and interpretation of forecasting methods rather than to generate the final experimental design itself.

AI tools were also used in connection with the development of the sentiment-aware components of the thesis, especially when refining explanations related to sentiment preprocessing, lagged feature construction, and the structure of the transparent Momentum Fear and Greed Index (MFGI). In these cases, the tools were used to clarify how technical ideas could be expressed more clearly and how specific implementation choices could be described in academically appropriate language.

In addition to technical support, AI tools were used during the writing process for revision and clarity improvement. These revisions included refining transitions between sections, improving sentence structure, strengthening explanations of forecasting concepts, and reorganizing drafted material into a clearer thesis structure. This support was used to improve readability and presentation rather than to replace the author’s own research, implementation, or analysis.

Throughout the writing process, grammar and proofreading tools were also used to identify punctuation errors, spelling mistakes, and minor grammatical issues. Any suggested revisions were reviewed individually before being accepted, and their use was limited to improving surface-level writing quality rather than generating substantive research content.

All outputs suggested by AI systems were treated only as advisory feedback and were reviewed, revised, and verified before being incorporated into either the implementation or the written thesis. All final decisions regarding the content of this thesis were made by the author.

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